SUSTech Southern University of Science and Technology

EE104 Fundamentals of Electric Circuits (Lesson 07)

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南方科技大学-工学院-电子与电气工程系-PN Lab

Mid-term exam

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- Important updateds

Chapters: chapters 1-7 **Time: 16:20 – 18:20** Nov.9th Saturday **Exam:** closed-book in English.

Location: The 3rd teaching building, and rooms can be found in the table below.

Tools: answer sheets and stretch paper will be provided. You will need calculators, pens and student ID cards. No phones, no smart watches, no books or any other documents related to circuits.

Key info. 1: One A4 cheating paper is allowed (both side, hand writing or printing).

Key info. 2: No make-up or online exams.

Key info. 3: SUBMIT all the exam paper, answer sheets, scratch papers, your cheating paper. **IT IS YOUR DUTY** to make sure all these documents are safely submitted to the TA at the end of the exam. IF a student loses one of these documents, he will be considered as **CHEATING** and score will be **ZERO**.

Class	1	2	3	4
Rooms	Rm105	Rm. 106	Rm. 107	Rm. 108
教室	三教105	三教106	三教107	三教108

Review: Lesson 6 (a)

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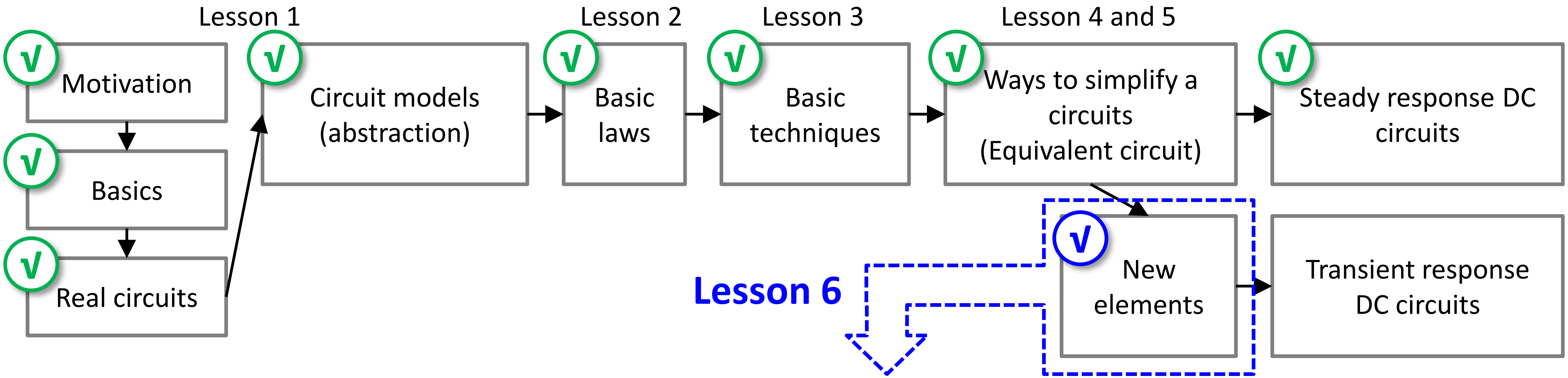
- EE104 road map



Capacitors and inductors

Review: Lesson 6 (b)

- Capacitors and inductors

a) Resistors, capacitors and inductors are all linear passive elements.

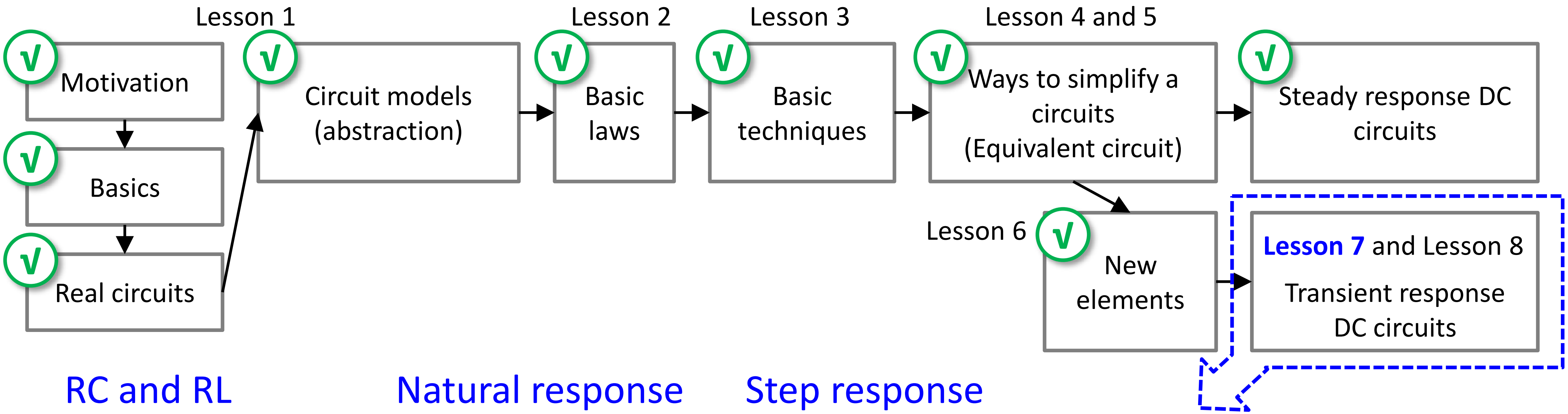
b) While ideal resistors consume energy, ideal capacitors and inductors store or release the energy without dissipation.

c) When capacitors and inductors are in steady state, they can be analyzed using equations shown in the right table.

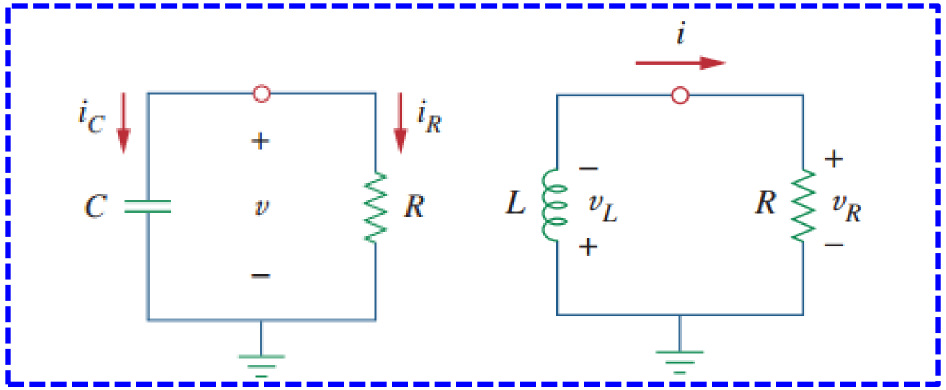
Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

Abstract of Lesson 7

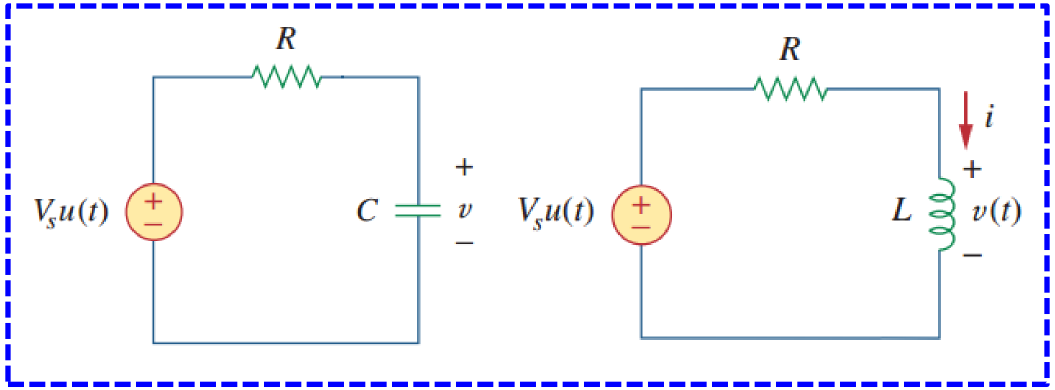
- Integrators and differentiators



RC and RL Natural response



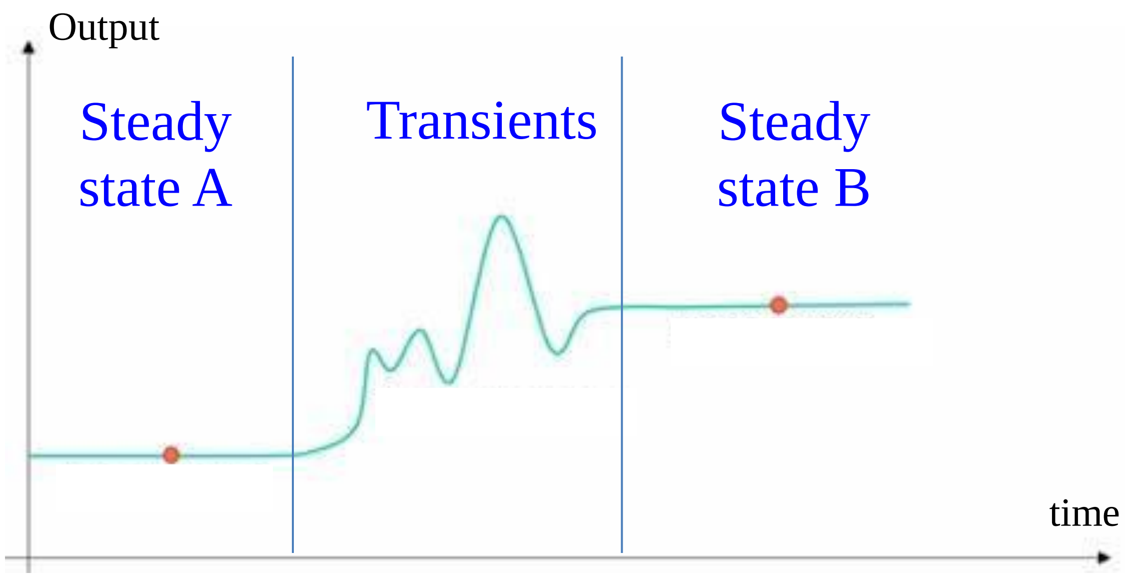
Step response



Lesson 7 First-order circuits

7.1 Introduction (a)

- Steady-state response and transient response



a) A circuit having constant sources is said to be in steady state if the currents and the voltages do not change with time. (circuits with currents and voltages having constant amplitude and constant frequency sinusoidal functions are also in a steady state.) Otherwise you can consider the circuits in a transient state.

b) The steady-state response refers to how a circuits responds to constant sources.

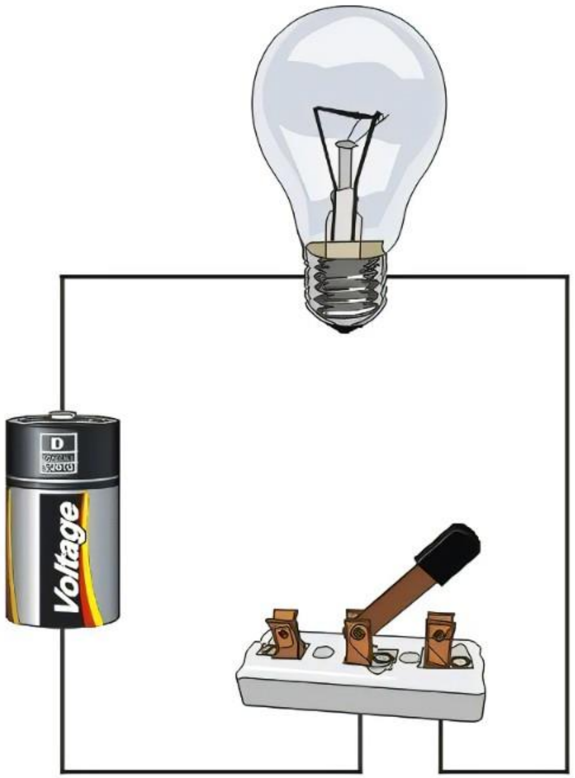
c) The transient response is the way the circuit responds to energies stored in energy-storage elements, such as capacitors and inductors.

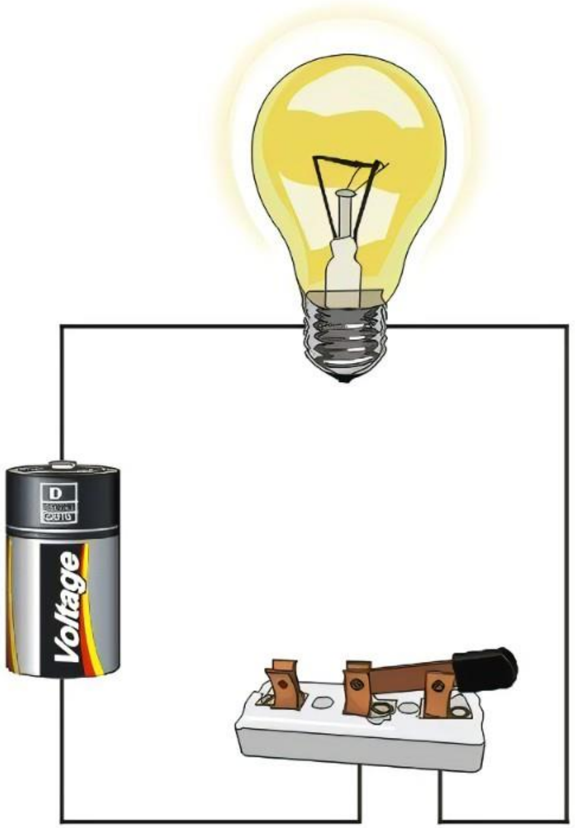
d) You can also consider the transient response is the response that happens when a circuit move from on steady state to another.

7.1 Introduction (b)

- Transients are important!

Transients





Steady state A

Steady state B

7.1 Introduction (c)

- Transients are important.

7.1 Introduction (d)

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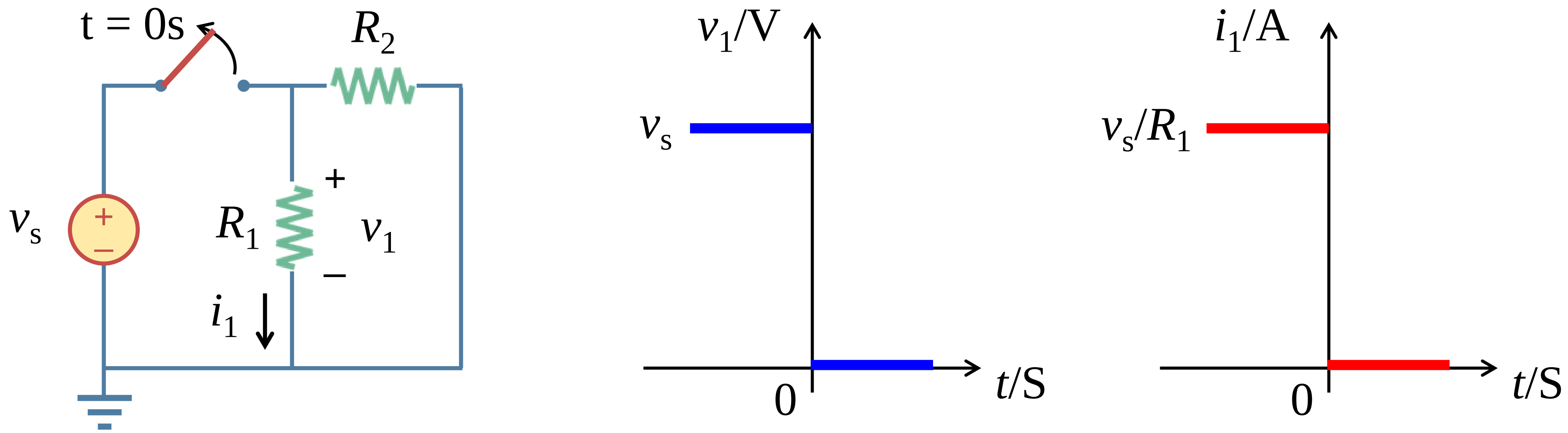
- Resistor transients



a) With ideal switches, there is no transition (过渡态) for resistor networks between ON (state 1) and OFF (state 2). The voltage and the current can both change abruptly (突变).

b) This is because resistors do not store the energy.

7.1 Introduction (e)

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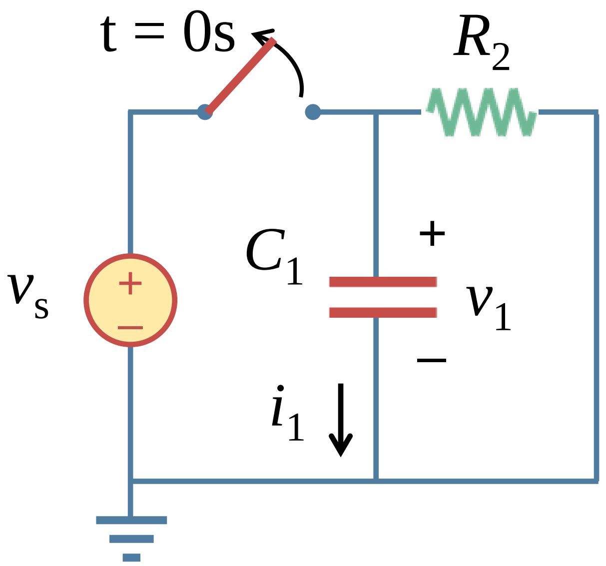
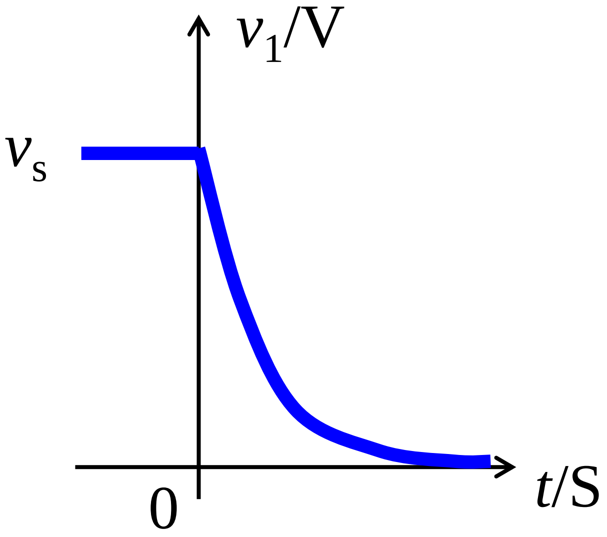
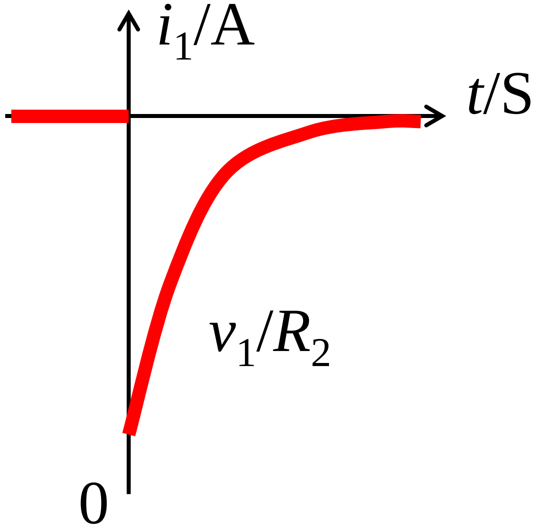
- Capacitor and inductor transients: discharging (放电)







a) However, there is transition (过渡态) for capacitors and inductors between initial (初态) and final states (终态), even with ideal switches, as the voltage across a capacitor and the current through an inductor cannot change abruptly (突变).

b) For capacitors ***i = Cdv/dt*** & for inductors ***v = Ldi/dt***

7.1 Introduction (f)

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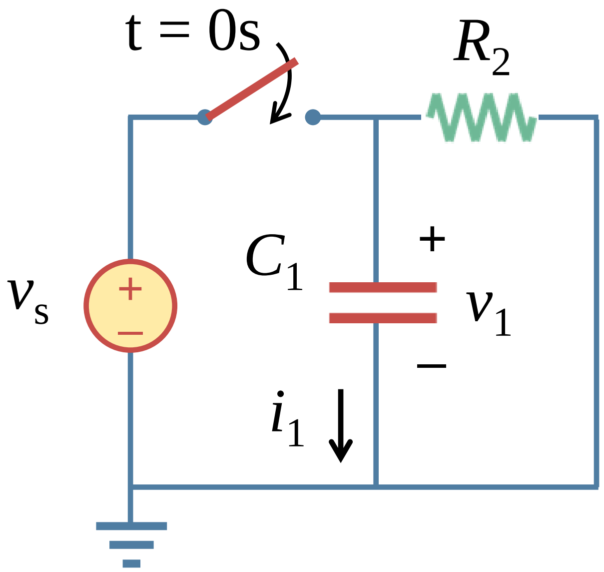
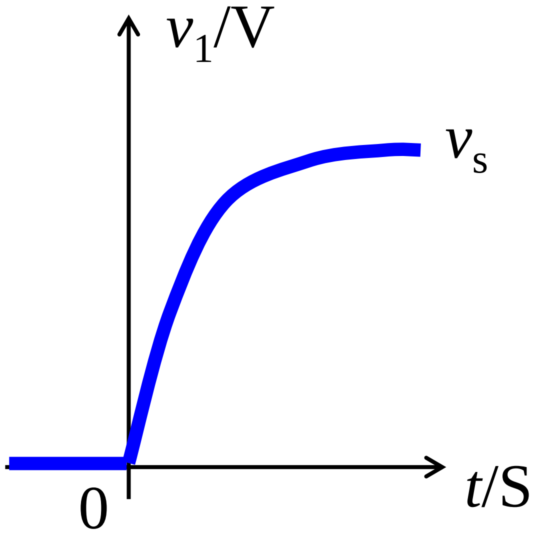
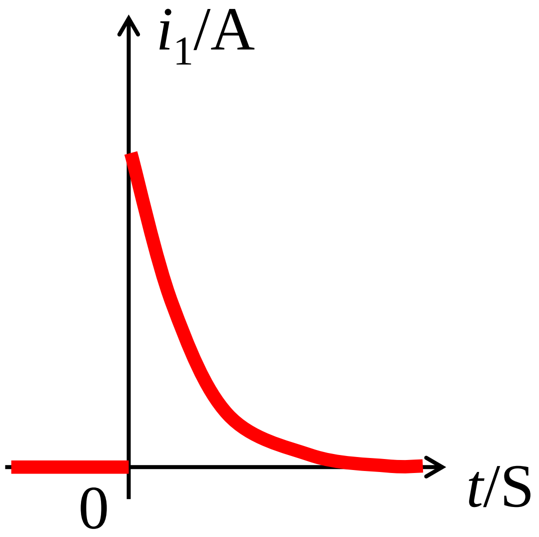
- Capacitor and inductor transients: charging (充电)





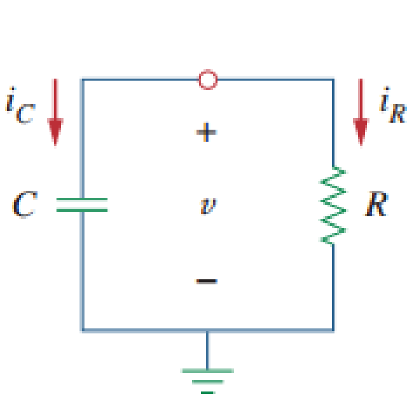
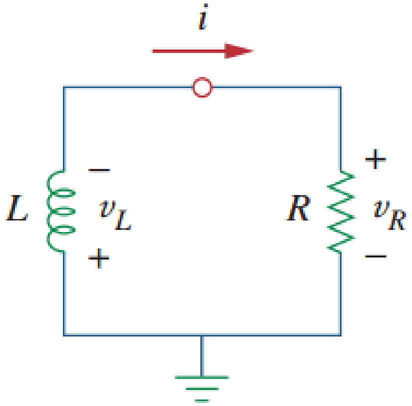
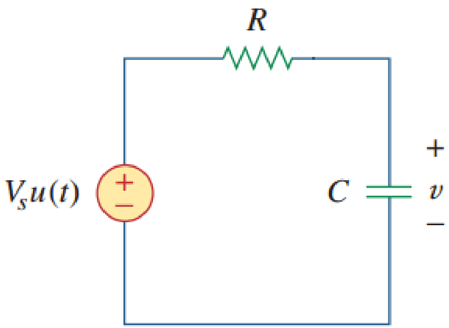
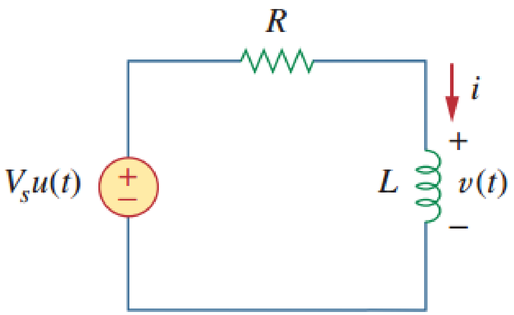


a) However, there is transition (过渡态) for capacitors and inductors between initial (初态) and final states (终态), even with ideal switches, as the voltage across a capacitor and the current through an inductor cannot change abruptly (突变).

b) For capacitors ***i* = *Cdv/dt*** & for inductors ***v* = *Ldi/dt***

7.1 Introduction (g)

- Transient behaviors of capacitors and inductors

	RC circuits	RL circuits
Source-free circuits		
Step response		

a) We use Kirchhoff's laws to analyze RC and RL circuits as we did for resistive circuits. The difference is that applying Kirchhoff's laws to purely resistive networks results in algebraic equations, while applying the laws to RC and RL circuits produces first-order differential equations.

b) Hence the circuits are known as first-order circuits. A first-order circuit is characterized by a first-order differential equation.

7.1 Introduction (h)

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- Quick review of first-order (linear) differential equations

When $a(x), b(x)$ *are constant and denote their rato* $\frac{a}{b} = \tau$, we have

case 1: $y'(t) + y(t)/\tau = 0$, with solution

$$y(t) = y(0)e^{-\frac{t}{\tau}}$$

case 2: $y'(t) + y(t)/\tau = C$, with solution

$$y(t) = y(\infty) + (y(0) - y(\infty))e^{-t/\tau}, \quad t \geq 0,$$

where, $y(\infty) = \tau C$

Lesson 7 First-order circuits

7.2 Source-free RC circuits (a)

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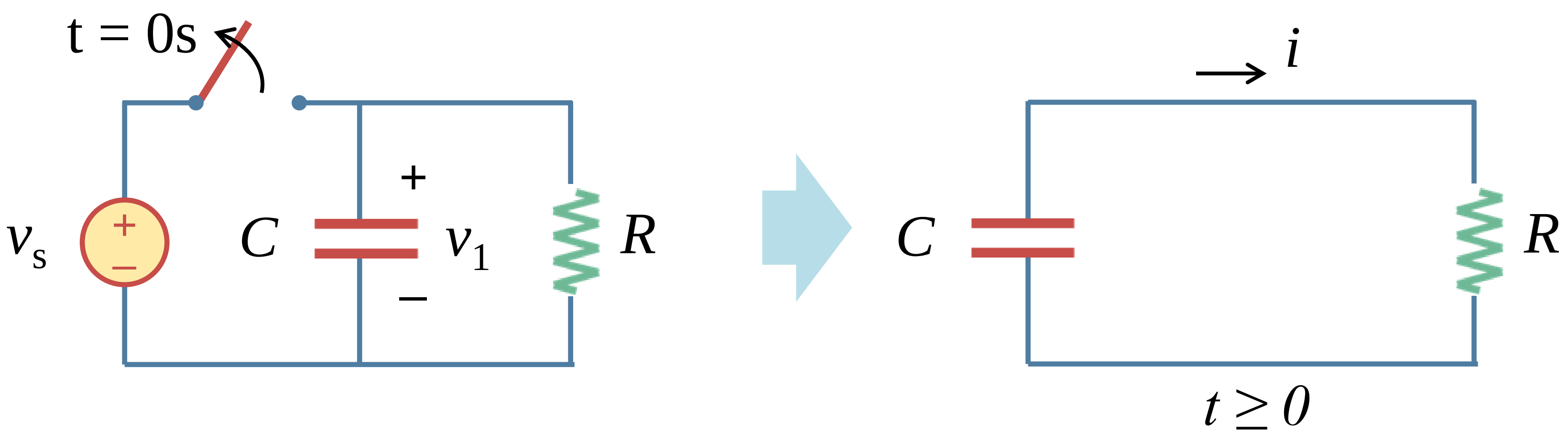
- A source-free RC circuits (discharging):



a) A source-free RC circuit occurs when the connected power source to the circuit is disconnected. The energy stored in the capacitor starts to dissipate as heat on the resistor, representing the discharging process of a capacitor with a resistive load.

7.2 Source-free RC circuits (b)

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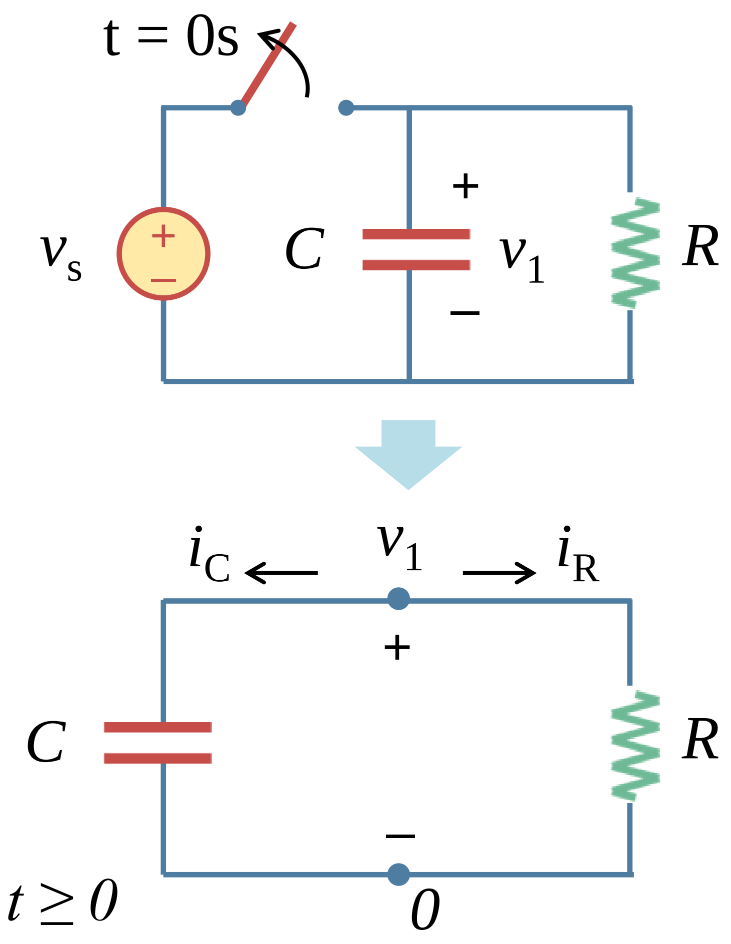
- A source-free RC circuits (discharging):



a) Using KCL at the top node:

$$C\frac{dv}{dt}+\frac{v}{R}=0 \rightarrow \frac{dv}{dt}+\frac{v}{RC}=0$$

b) The solution:

$$\boxed{v(t) = V_0 e^{-t/RC} = V_0\, e^{-t/\tau}}$$

c) Where $\tau = RC$ is called time constant (时间常数) of the RC circuit; V_0 is the initial voltage across the capacitor.

d) Note that the resistor and capacitor may be the equivalent resistor and equivalent capacitor of complex resistive and capacitive networks.

7.2 Source-free RC circuits (c)

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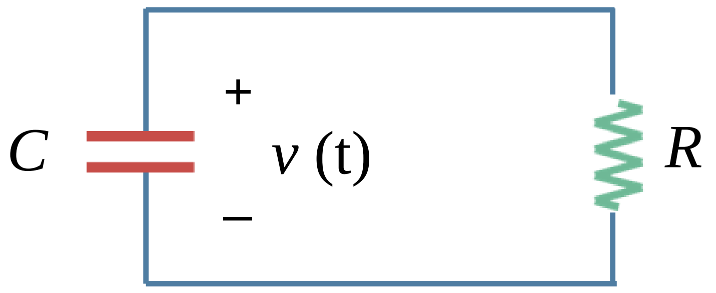
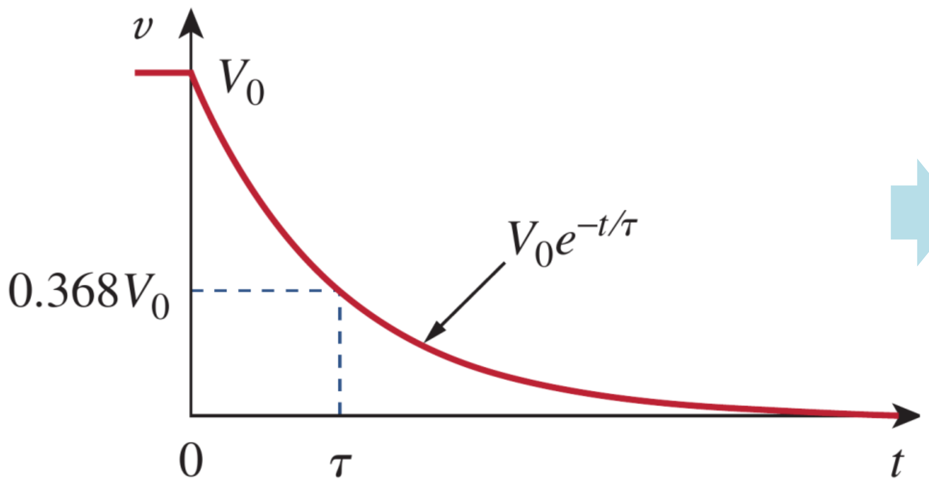
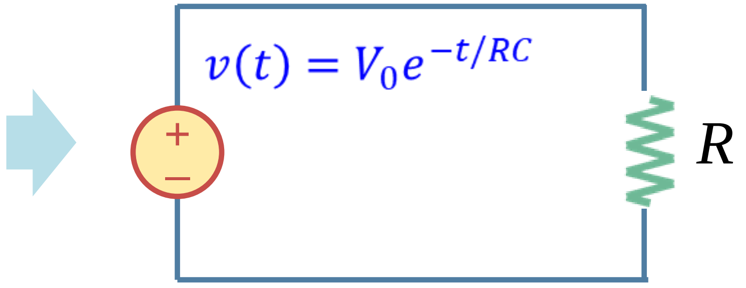
- Natural response (自由响应) of charged capacitors:

Source-free RC circuits



$$v(t) = V_0 e^{-t/RC} = V_0\ e^{-t/\tau}$$





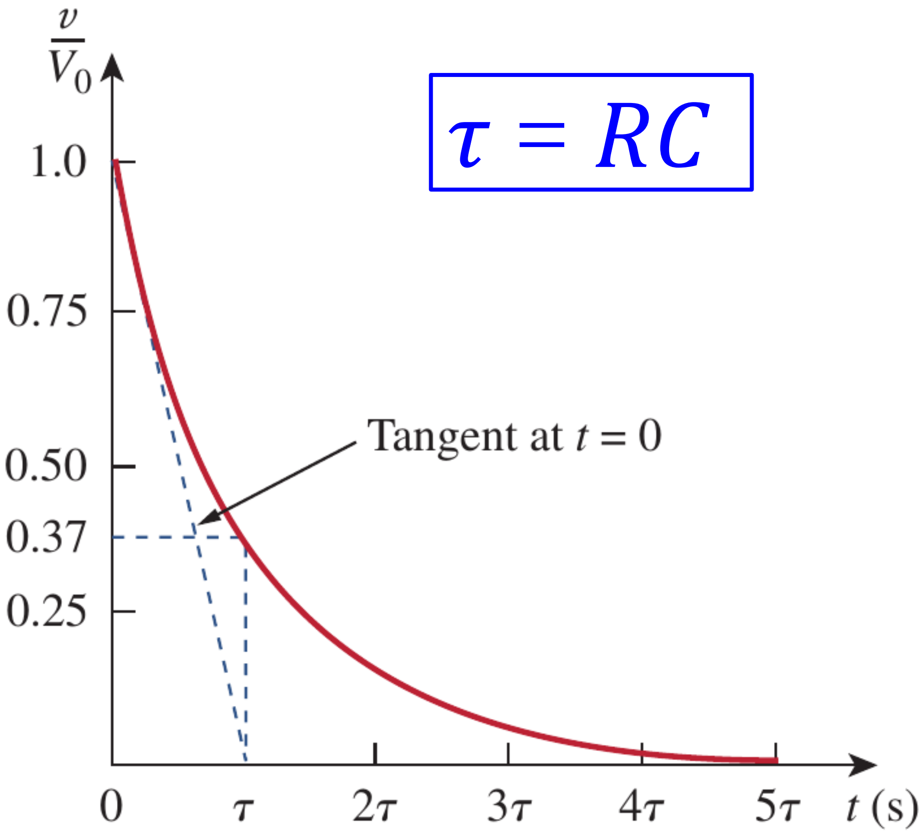
a) Given the fact that there is no external sources, the response $v(t)$ is also called the natural response (自由响应).

b) It could be considered as a voltage source, whose voltage decays exponentially with time.

c) The speed of decay is determined by the time constant τ.

7.2 Source-free RC circuits (d)

- Time constant (τ):

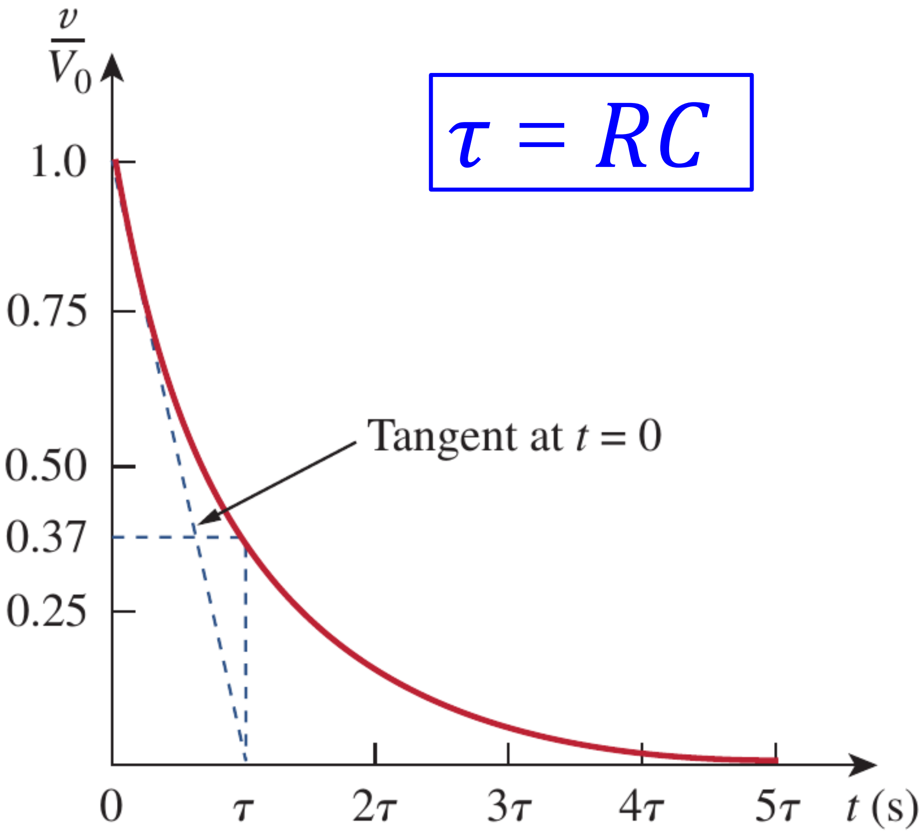


$$\boxed{\tau = RC}$$

t	$v(t)/V_0$
τ	0.36788
2τ	0.13534
3τ	0.04979
4τ	0.01832
5τ	0.00674

a) The time constant (τ) of a circuit is the time required for the response to decay to a factor of 1/e or 36.8% of its initial value. At any rate, whether the time constant is small or large, the circuit reaches steady state in five time constants.

- Time constant (τ):



$$\boxed{\tau = RC}$$

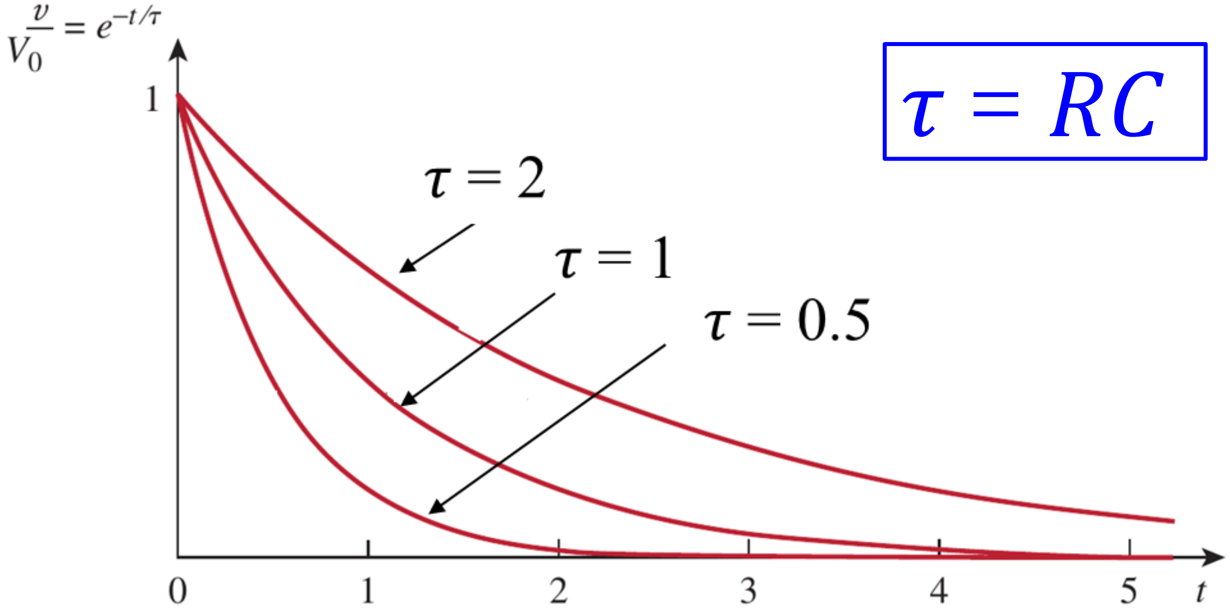
$$\frac{d}{dt}\left(\frac{v}{V_0}\right)\bigg|_{t=0} = -\frac{1}{\tau}e^{-t/\tau}\bigg|_{t=0} = -\frac{1}{\tau}$$

b) The time constant (τ) is the initial rate of decay: or the time taken for v/V_0 to decay from unity to zero, assuming a constant rate of decay. This initial slope interpretation of the time constant is often used in the laboratory to find t graphically from the response curve displayed on an oscilloscope.

7.2 Source-free RC circuits (f)

- Time constant (τ) represents the speed of decay:

a) A circuit with a small τ gives a fast response in that it reaches the steady state (or final state) quickly due to quick dissipation of energy stored, yet a circuit with a large τ gives a slow response as it takes longer to reach steady state.



$$\boxed{\tau = RC}$$

b) The smaller the τ, the more rapidly the voltage decreases, that is, the faster the response.

c) At any rate, whether the τ is small or large, the circuit reaches steady state in five time constants.

7.2 Source-free RC circuits (g)

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- Current and energy response:

$$\boxed{\tau = RC}$$

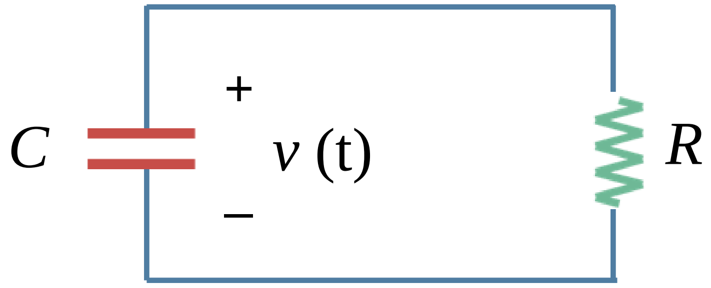
a) The current:

$$\boxed{i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}}$$

b) The power dissipated in the resistor:

$$\boxed{p(t) = v i_R = \frac{V_0^2}{R} e^{-2t/\tau}}$$

Source-free RC circuits



$$v(t) = V_0 e^{-t/RC} = V_0\, e^{-t/\tau}$$

c) The energy dissipated by the resistor up to time *t* is:

$$\boxed{w_R(t) = \int_0^t p(\lambda)\,d\lambda = \frac{V_0^2}{R}\int_0^t e^{-2\lambda/\tau} d\lambda = -\frac{1}{2}\tau \cdot \frac{V_0^2}{R} e^{-2\lambda/\tau}\Big|_0^t = \frac{1}{2} C V_0^2 \left(1 - e^{-2t/\tau}\right)}$$

7.3 Examples (a)

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- General procedures to analyze source-free RC circuits:

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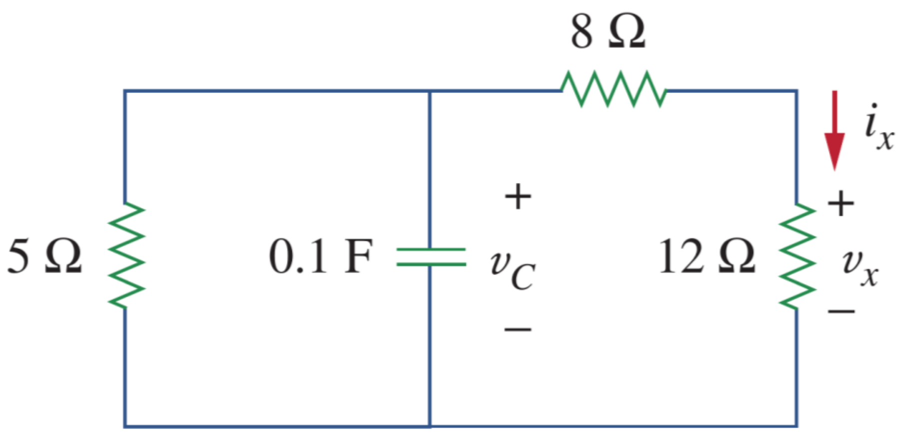
As the equation $v_c(t) = V_0 e^{-t/\tau}$ shows, to determine the response of a source-free RC circuit, we only need to know

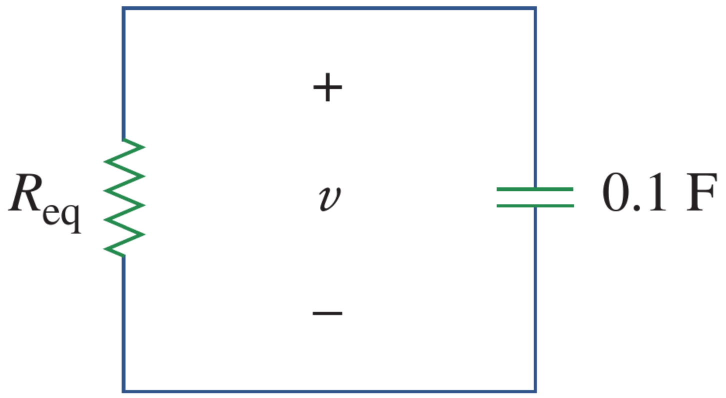
1. The **initial voltage** $\boldsymbol{v(t) = V_0}$ on capacitor. The initial voltage can be obtained from the steady analysis of the circuit (treat capacitor as open circuit)

2. The **time constant** $\boldsymbol{\tau = RC}$. In finding the time constant $\boldsymbol{\tau}$, one may need to use the Thevenin theorem to find out the equivalent resistance seen by the capacitor (the circuit seen by the capacitor may contain more than one resistors and dependent sources).

- Example 7.01:

Example 7.01: In Fig. 7.5, let v_C (0) = 15 V. Find v_C, v_x, and i_x for t > 0.





Solution:

$$R_{eq} = \frac{20 \times 5}{20 + 5} = 4\ \Omega \qquad \tau = R_{eq}C = 4(0.1) = 0.4\ \mathrm{s}$$

$$v_C = v = v(0)e^{-t/\tau} = 15e^{-t/0.4}\ \mathrm{V}$$

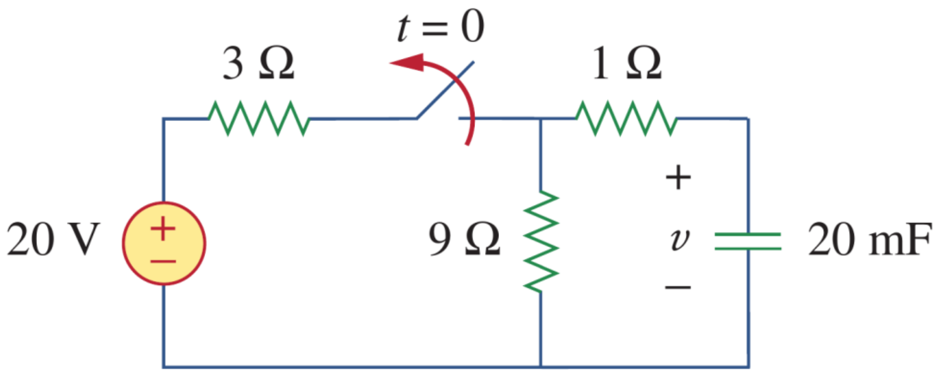
$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t}\ \mathrm{V}$$

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t}\ \mathrm{A}$$

7.3 Examples (c)

- Example 7.02:

Example 7.02: The switch in the following circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t > 0$. Calculate the initial energy stored in the capacitor.



Solution:

(1) We can easily find out the initial voltage is: $v(0) = 20 \times 9/(3+9) = 15\,V$

(2) After the switch is opened, the circuit at the right side of the switch becomes a source-free RC circuit. The Thevenin equivalent resistance seen by the capacitor is:

$$R_{eq} = 1 + 9 = 10\Omega$$

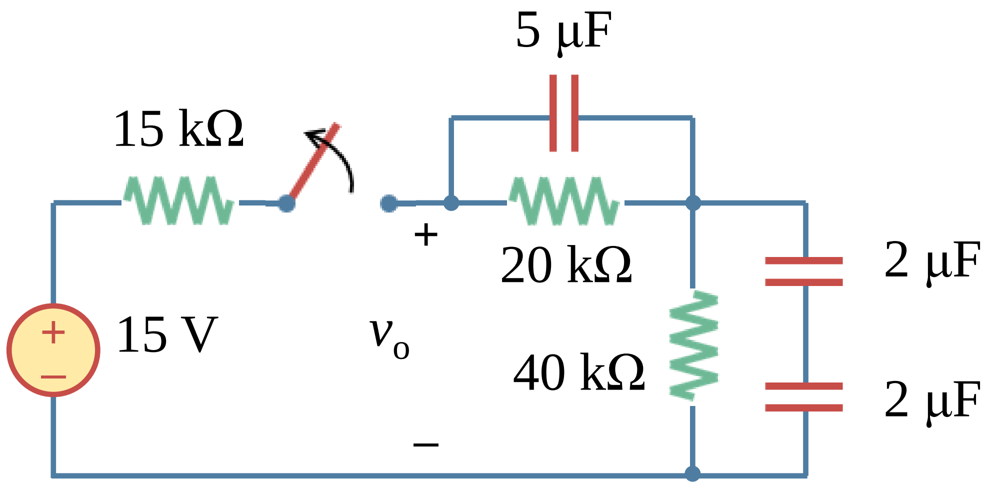
(3) the time constant of the RC circuit is: $\tau = R_{eq}C = 10 \cdot 20 = 200\text{ ms} = 0.2\ s$

(4) So, the capacitor voltage is: $v(t) = 15e^{-5t}V$

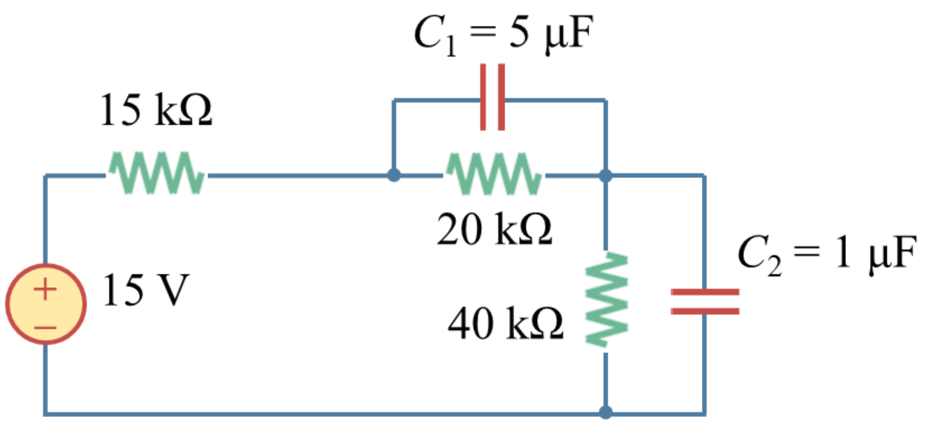
7.3 Examples (d)

- Example 7.03:

Example 7.03: The switch in the following circuit has been closed for a long time, and it is opened at *t=0*. (a) Find $v_o(t)$ for $t > 0$. (b) Calculate the percentage of energy consumed as compared to the initial energy stored in the capacitors at 60 mS.



Solution: **(1) to find $v_1(0)$ and $v_2(0)$**



$$\frac{1}{C_2} = \frac{1}{2\mu} + \frac{1}{2\mu} \Rightarrow C_2 = 1\ \mu\text{F}$$

$$v_1(0) = \frac{20}{15+20+40} \times 15 = 4\text{ V}$$

$$v_2(0) = \frac{40}{15+20+40} \times 15 = 8\text{ V}$$

7.3 Examples (e)

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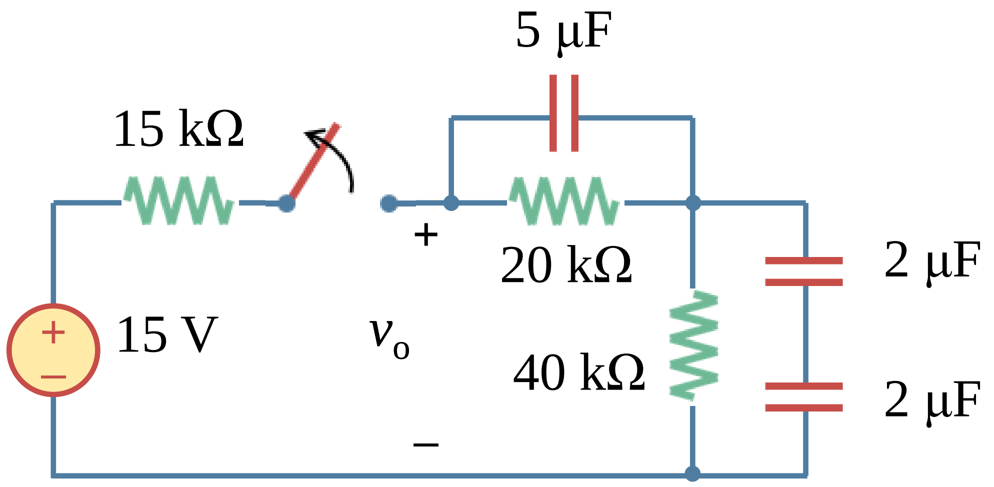
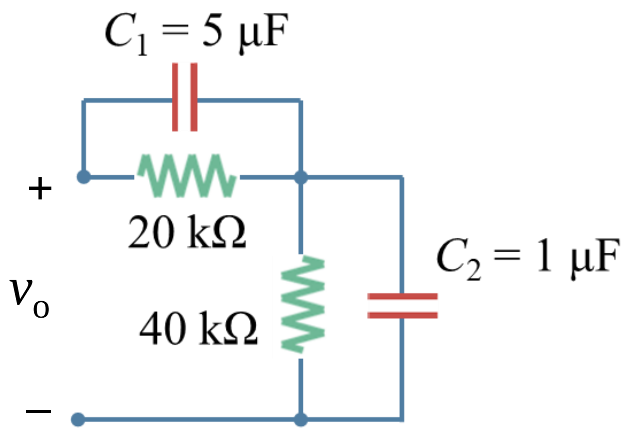
- Example 7.03:

Example 7.03: The switch in the following circuit has been closed for a long time, and it is opened at $t=0$. (a) Find $v_o(t)$ for $t > 0$. (b) Calculate the percentage of energy consumed as compared to the initial energy stored in the capacitors at 60 mS.



Solution: **(2) to find $v_1(t)$ and $v_2(t)$:**



$$\tau_1 = 20\text{ k} \times 5\ \mu = 0.1\text{ s} \rightarrow v_1 = 4e^{-t/\tau} = 4e^{-10t}\text{ V}$$

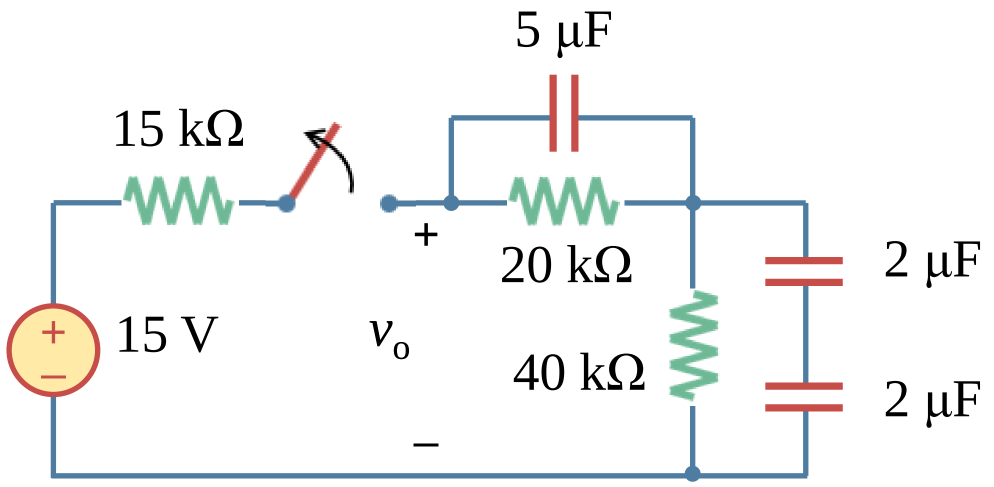
$$\tau_2 = 40\text{ k} \times 1\ \mu = 0.04\text{ s} \rightarrow v_2 = 8e^{-t/\tau} = 8e^{-25t}\text{ V}$$

$$v_o(t) = v_1(t) + v_2(t) = 4e^{-10t} + 8e^{-25t}\text{ V}$$

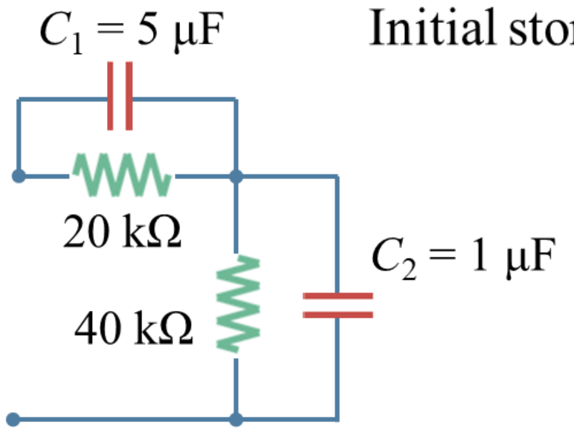
7.3 Examples (f)

- Example 7.03:

Example 7.03: The switch in the following circuit has been closed for a long time, and it is opened at $t=0$. (a) Find $v_o(t)$ for $t > 0$. (b) Calculate the percentage of energy consumed as compared to the initial energy stored in the capacitors at 60 mS.



Solution: **(3) to find w_1(t) and w_2(t) :**



Initial stored energy: $w(0) = w_1(0) + w_2(0) = \frac{1}{2}C_1[v_1(0)]^2 + \frac{1}{2}C_2[v_2(0)]^2 = 72\ \mu\text{J}$

Energy consumed: $w_1(60\text{ mS}) = \frac{1}{2}C_1[v_1(0)]^2\left(1 - e^{-2t/\tau}\right) \approx 28\ \mu\text{J}$ (70%)

$w_2(60\text{ mS}) = \frac{1}{2}C_2[v_2(0)]^2\left(1 - e^{-2t/\tau}\right) \approx 30\ \mu\text{J}$ (95%)

Percentage of Energy consumed: $(28 + 30)/72 \approx 80.5\%$

7.4 Source-free RL circuits (a)

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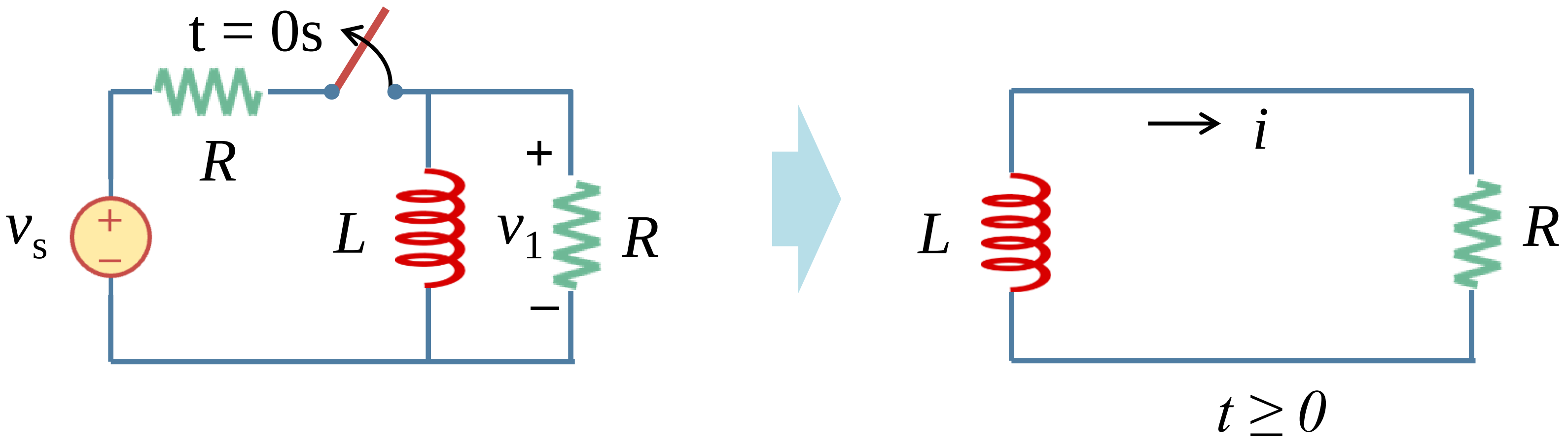
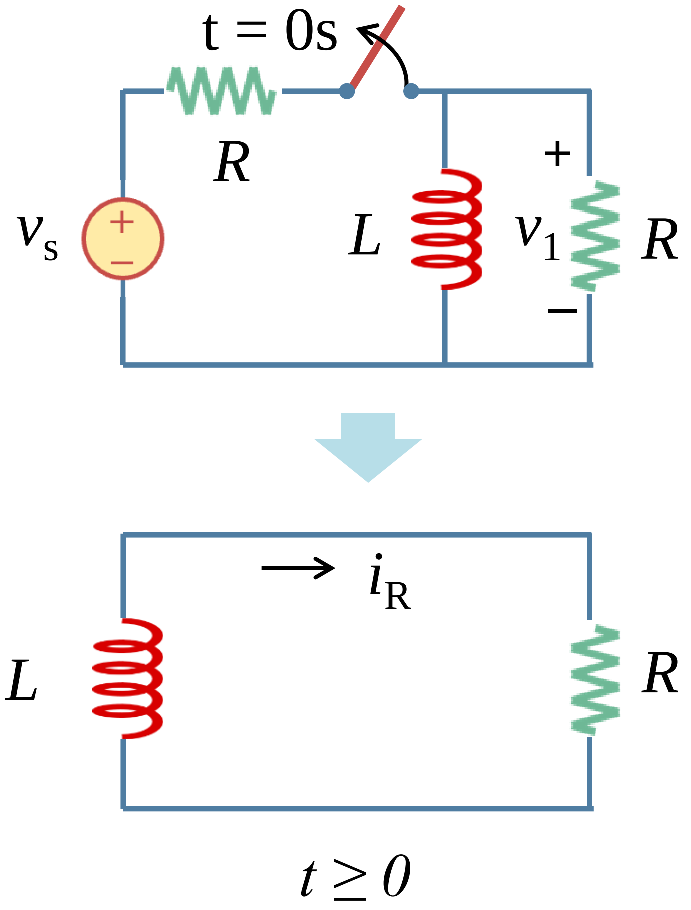
- A source-free RL circuits (discharging):



a) A source-free RL circuit occurs when the connected power source to the circuit is disconnected. The energy stored in the inductor starts to dissipate as heat on the resistor, representing the discharging process of a inductor with a resistive load.

- A source-free RL circuits (discharging):



a) Using KVL at the top node:

$$v_L + v_R = 0 \rightarrow L\frac{di}{dt} + Ri = 0 \rightarrow \frac{di}{dt} + \frac{R}{L}i = 0$$

b) The solution:

$$\boxed{i(t) = i(0)e^{-Rt/L} = i(0)e^{-t/\tau}}$$

c) Where $\tau = L/R$ is called time constant (时间常数) of the RL circuit; V_0 is the initial voltage across the capacitor.

d) Note that the resistor and inductor may be the equivalent resistor and equivalent capacitor of complex resistive and capacitive networks.

7.4 Source-free RL circuits (c)

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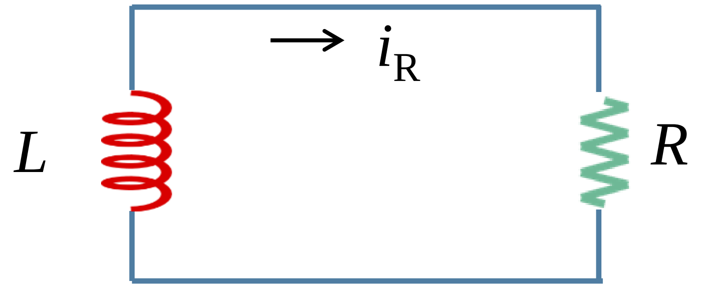
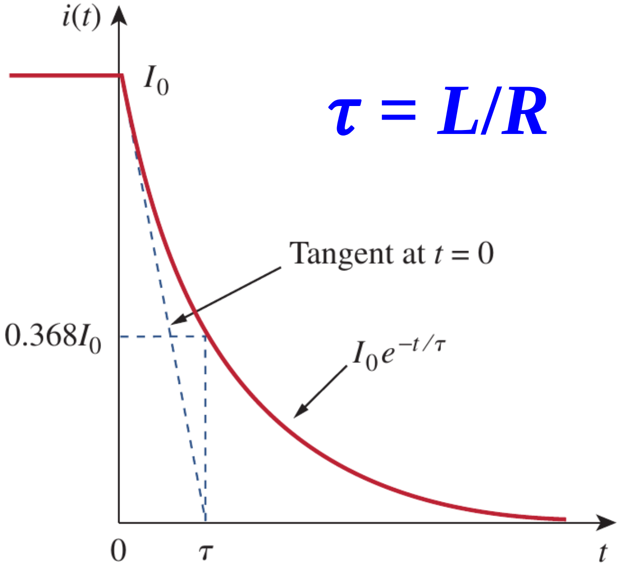
- Natural response (自 由响应) of charged inductors:

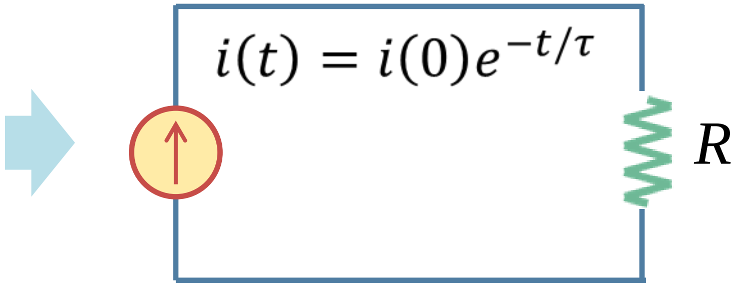
Source-free RL circuits



$$i(t) = i(0)e^{-Rt/L} = i(0)e^{-t/\tau}$$



$\tau = L/R$



a) Given the fact that there is no external sources, the response $i(t)$ is also called the natural response (自 由响应).

b) It could be considered as a current source, whose voltage decays exponentially with time.

c) The speed of decay is determined by the time constant τ.

7.4 Source-free RL circuits (d)

- Voltage and energy response:

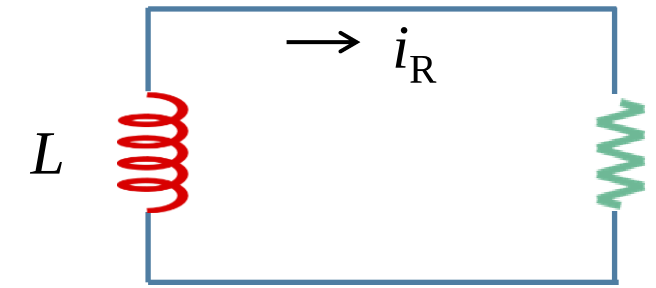
a) The voltage:

$$\boxed{v_R(t) = i(t)R = I_0 R e^{-t/\tau}}$$

b) The power dissipated in the resistor:

$$\boxed{p(t) = v_R i = I_0^2 R e^{-2t/\tau}}$$

$\boldsymbol{\tau = L/R}$



$$i(t) = i(0)e^{-Rt/L} = i(0)e^{-t/\tau}$$

c) The energy dissipated by the resistor up to time t is:

$$\boxed{w_R(t) = \int_0^t p(\lambda)\,d\lambda = I_0^2 R \int_0^t e^{-2\lambda/\tau}\,d\lambda = -\frac{1}{2}\tau \cdot I_0^2 R e^{-2\lambda/\tau}\Big|_0^t = \frac{1}{2} L I_0^2 \left(1 - e^{-2t/\tau}\right)}$$

7.4 Source-free RL circuits (e)

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- General procedures to analyze source-free RC circuits:

As the equation $i(t) = i(0)e^{-\frac{t}{\tau}} = i(0)e^{-\frac{Rt}{L}}$ shows, to determine the response of a source-free RC circuit, we only need to know

1. The **initial current *i(t)* = i_0** through the inductor. The initial current can be obtained from the steady analysis of the circuit (treat capacitor as short circuit).

2. The **time constant $\tau = L/R$**. In finding the time constant $\boldsymbol{\tau}$, one may need to use the Thevenin theorem to find out the equivalent resistance seen by the inductor (the circuit seen by the capacitor may contain more than one resistors and dependent sources).

7.4 Source-free RL circuits (f)

7.5 Examples (a)

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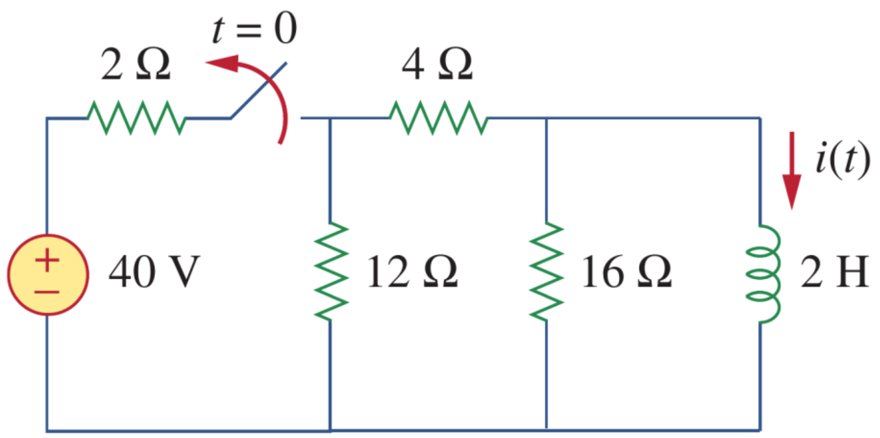
- Example 7.04:

Example 7.04: The switch in the circuit has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

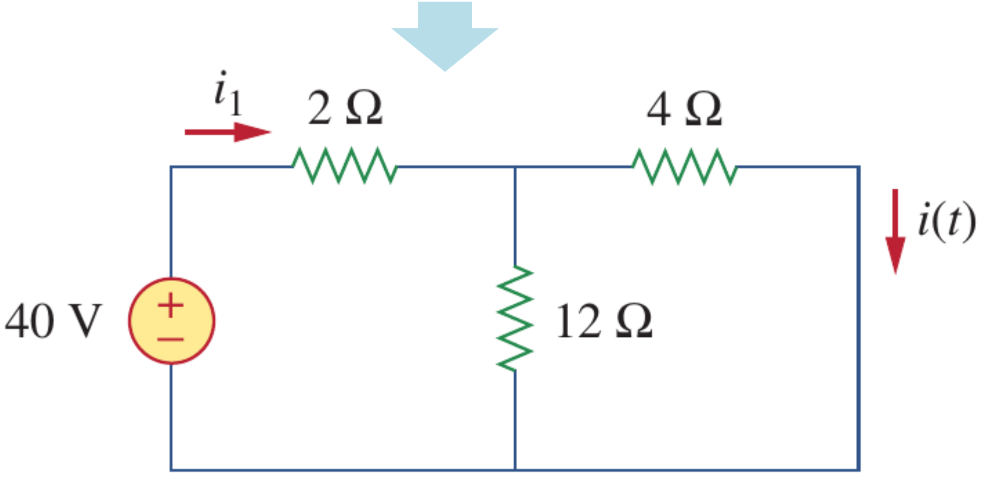


Solution:

(1) Find the initial current $i(0)$

$$\frac{4 \times 12}{4 + 12} = 3\ \Omega \quad \Rightarrow \quad i_1 = \frac{40}{2 + 3} = 8\ \text{A}$$

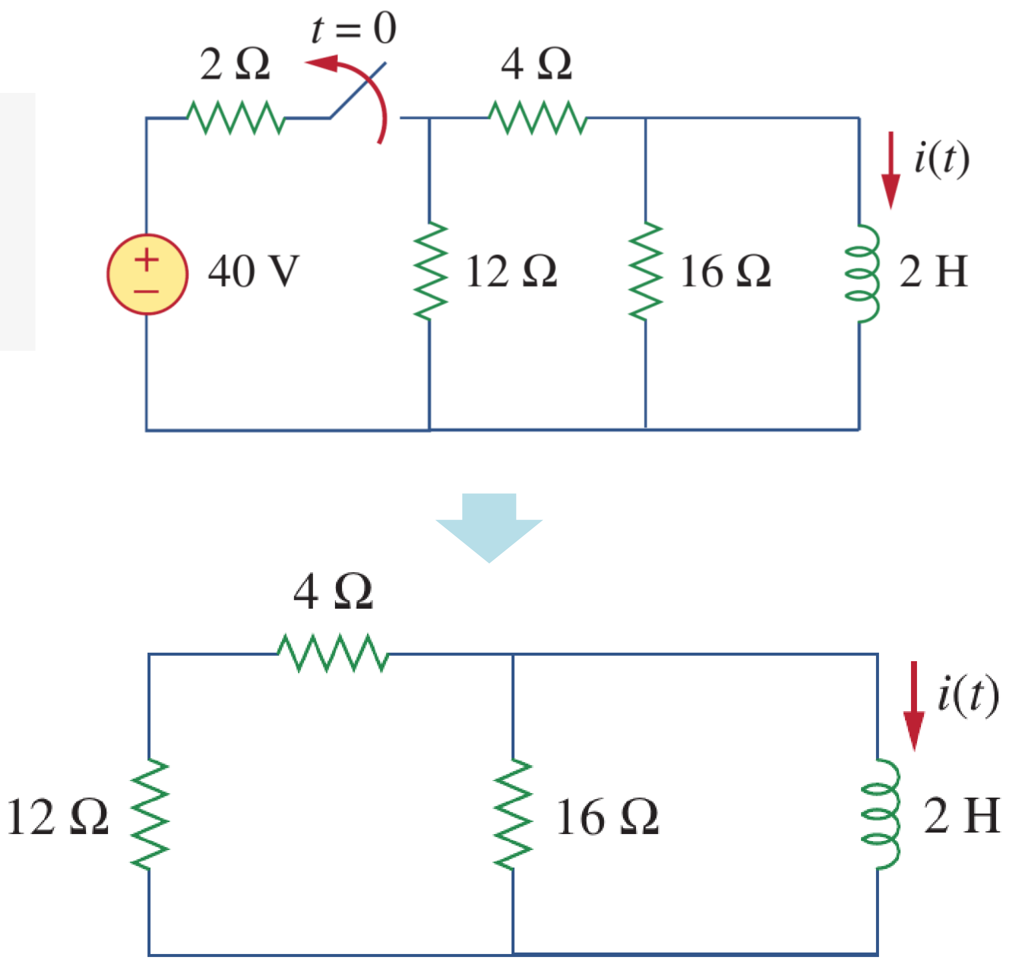
$$i(t) = \frac{12}{12 + 4} i_1 = 6\ \text{A}, \quad t = 0$$



7.5 Examples (b)

- Example 7.04:

Example 7.04: The switch in the circuit has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.



Solution:

(2) Find the time constant → The equivalent resistance

$$R_{\text{eq}} = (12 + 4) \parallel 16 = 8\ \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{2}{8} = \frac{1}{4}\ \text{s}$$

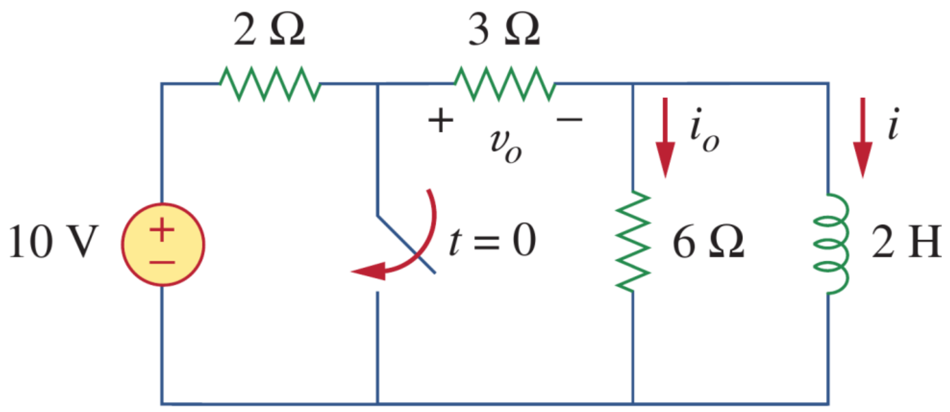
(3) with i_0 and τ, now you can find $i(t)$

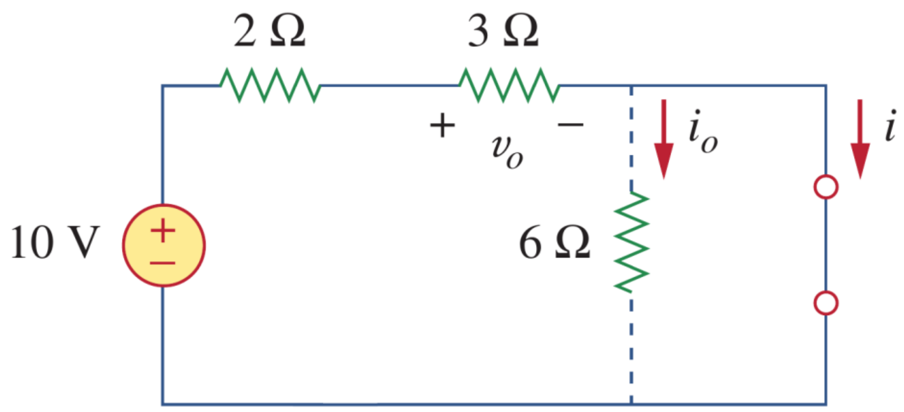
$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t}\ \text{A}$$

7.5 Examples (c)

- Example 7.05:

Example 7.05: find i_o, v_o, and i for all time, assuming that the switch was open for a long time.





Solution: (1) Find $i(0)$ when the switch is open

The switch is open when $t < 0$,

$$i(t) = \frac{10}{2+3} = 2\text{ A}, \qquad t < 0$$

$$v_o(t) = 3i(t) = 6\text{ V}, \qquad t < 0$$

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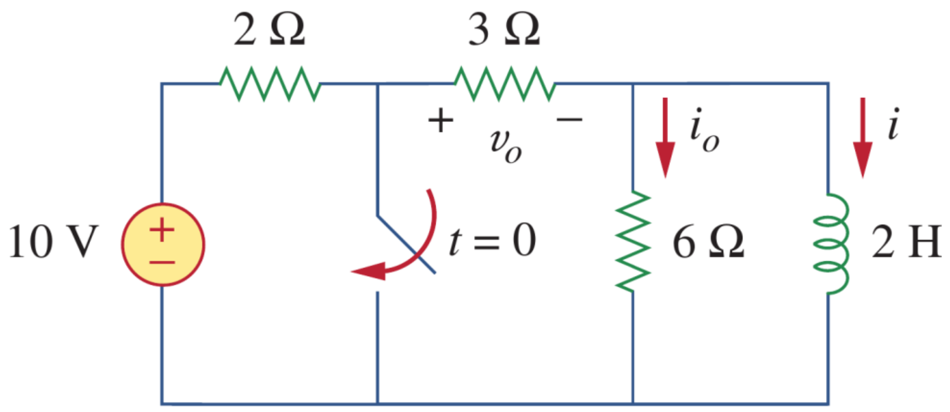
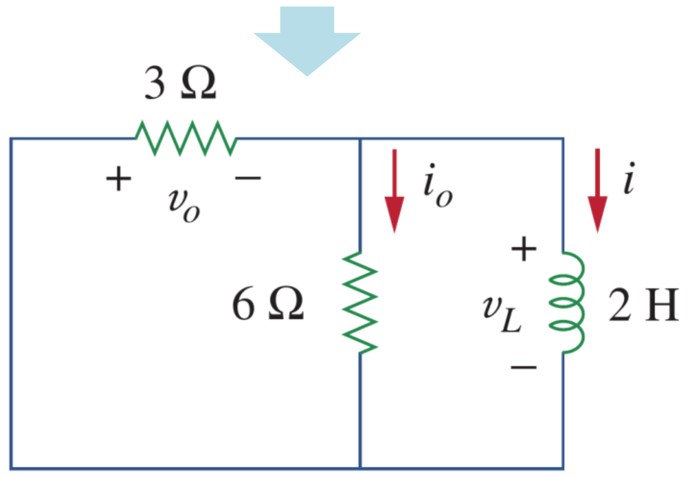
- Example 7.05:

Example 7.05: find i_o, v_o, and i for all time, assuming that the switch was open for a long time.





Solution: (2) The time constant → the equivalent resistance

$$R_{\text{Th}} = 3 \parallel 6 = 2\ \Omega \quad \Rightarrow \quad \tau = \frac{L}{R_{\text{Th}}} = 1\ \text{s}$$

(3) Now you can calculate other unknowns

$$i(t) = i(0)e^{-t/\tau} = 2e^{-t}\ \text{A}, \qquad t > 0$$

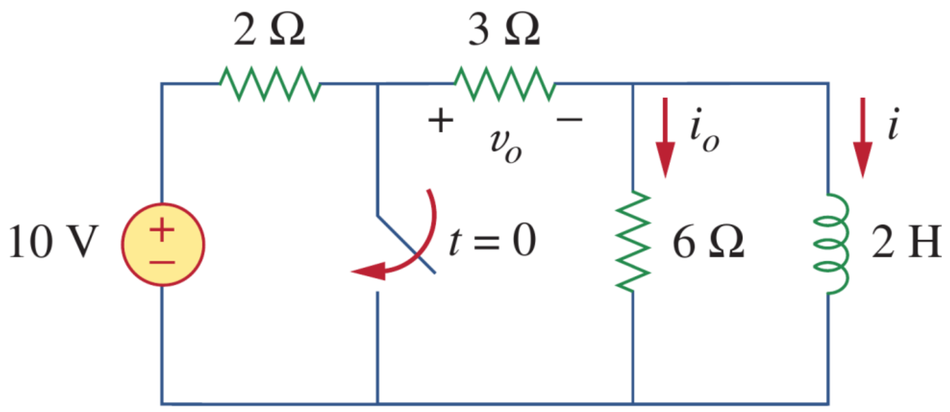
$$v_o(t) = -v_L = -L\frac{di}{dt} = -2(-2e^{-t}) = 4e^{-t}\ \text{V}, \qquad t > 0$$

$$i_o(t) = \frac{v_L}{6} = -\frac{2}{3}e^{-t}\ \text{A}, \qquad t > 0$$

7.5 Examples (e)

- Example 7.05:

Example 7.05: find i_o, v_o, and i for all time, assuming that the switch was open for a long time.

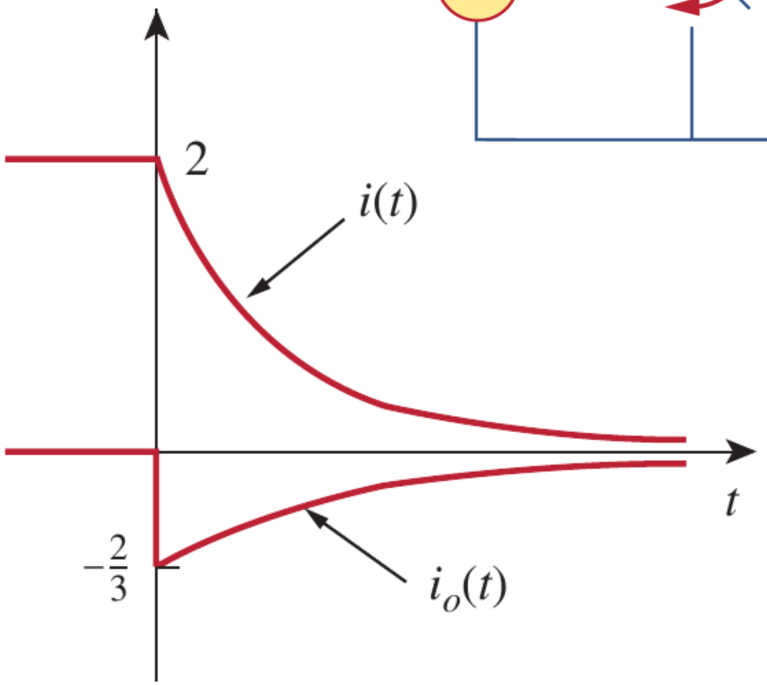


Solution:

$$i_o(t) = \begin{cases} 0\text{ A}, & t < 0 \\ -\dfrac{2}{3}e^{-t}\text{ A}, & t > 0 \end{cases},$$

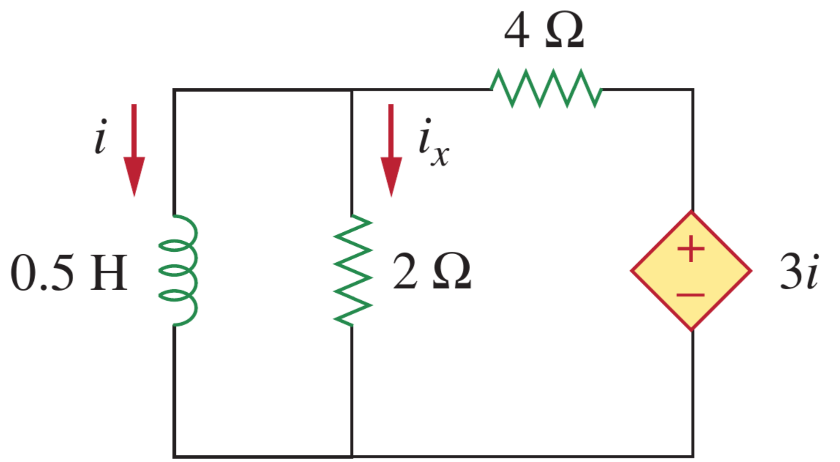
$$v_o(t) = \begin{cases} 6\text{ V}, & t < 0 \\ 4e^{-t}\text{ V}, & t > 0 \end{cases}$$

$$i(t) = \begin{cases} 2\text{ A}, & t < 0 \\ 2e^{-t}\text{ A}, & t \geq 0 \end{cases}$$



7.5 Examples (f)

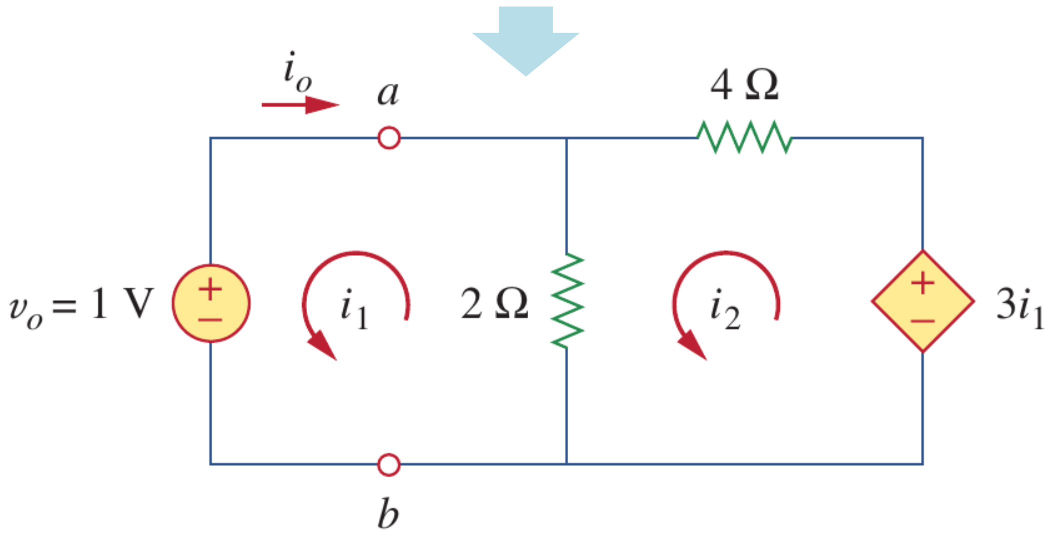
- Example 7.06:



Example 7.06: Assuming that $i(0) = 10$ A, calculate $i(t)$ and $i_x(t)$ in the following circuit for $t > 0$.

Solution: (1) the initial current is $i(0) = 10$ A

(2) the time constant → The equivalent resistance



$$2(i_1 - i_2) + 1 = 0 \quad \Rightarrow \quad i_1 - i_2 = -\frac{1}{2}$$

$$6i_2 - 2i_1 - 3i_1 = 0 \quad \Rightarrow \quad i_2 = \frac{5}{6}i_1$$

$$i_1 = -3 \text{ A}, \qquad i_o = -i_1 = 3 \text{ A}$$

$$R_{eq} = R_{Th} = \frac{v_o}{i_o} = \frac{1}{3}\,\Omega \quad \Rightarrow \quad \tau = \frac{L}{R_{eq}} = \frac{\frac{1}{2}}{\frac{1}{3}} = \frac{3}{2} \text{ s}$$

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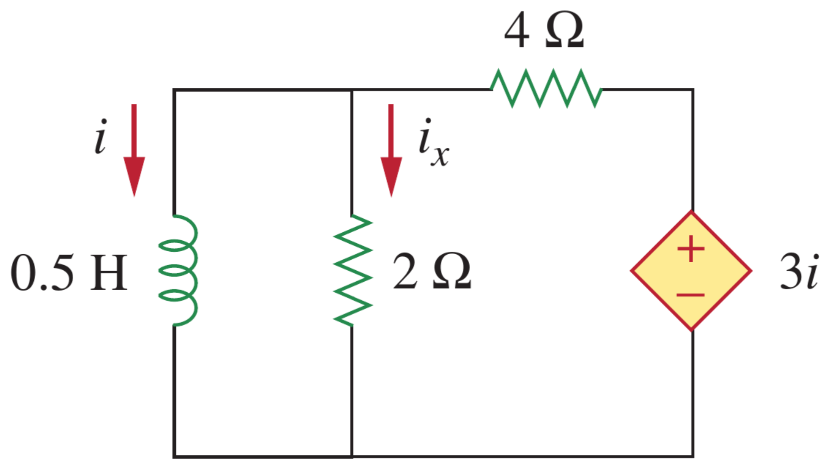
- Example 7.06:

Example 7.06: Assuming that $i(0) = 10$ A, calculate i(t) and i_x(t) in the following circuit.



Solution: (3) Now you have $i(0)$ and τ

$$i(t) = i(0)e^{-t/\tau} = 10e^{-(2/3)t} \text{ A}$$

(4) To calculate the i_x: the voltage across the inductor is

$$v = L\frac{di}{dt} = 0.5(10)\left(-\frac{2}{3}\right)e^{-(2/3)t} = -\frac{10}{3}e^{-(2/3)t} \text{ V}$$

$$i_x(t) = \frac{v}{2} = -1.6667e^{-(2/3)t} \text{ A.}$$

Lesson 7 First-order circuits

7.6 Singularity Functions (a)

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- Singularity Functions (奇异函数):

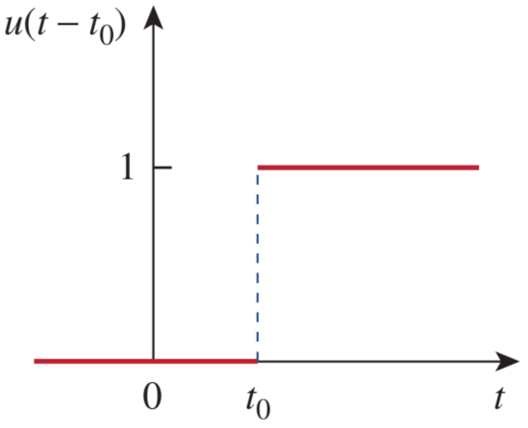
a) Singularity functions are functions that either are discontinuous or have discontinuous derivatives, which as good approximations to the switching signals that arise in circuits with switching operations.

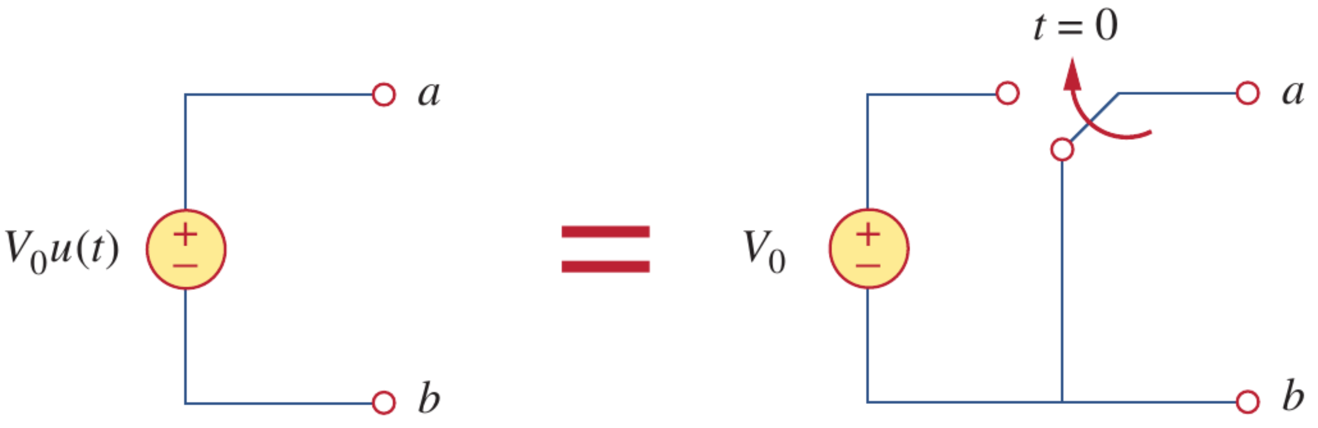
b) A basic understanding of singularity functions will help us make sense of the response of first-order circuits to a sudden application of an independent dc voltage or current source (charging), They are helpful in the neat, compact description of some circuit phenomena, especially the step response of RC or RL circuits.

c) The three most widely used singularity functions in circuit analysis are the unit step (单位阶跃), the unit impulse (单位冲激), and the unit ramp functions (单位斜坡).

7.6 Singularity Functions (b)

- Unit step function (单位阶跃函数):





$$v(t) = V_0 u(t - t_0)$$

$$u(t-t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases} \qquad v(t) = \begin{cases} 0, & t < t_0 \\ V_0, & t > t_0 \end{cases} \qquad u(t+t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$

7.6 Singularity Functions (c)

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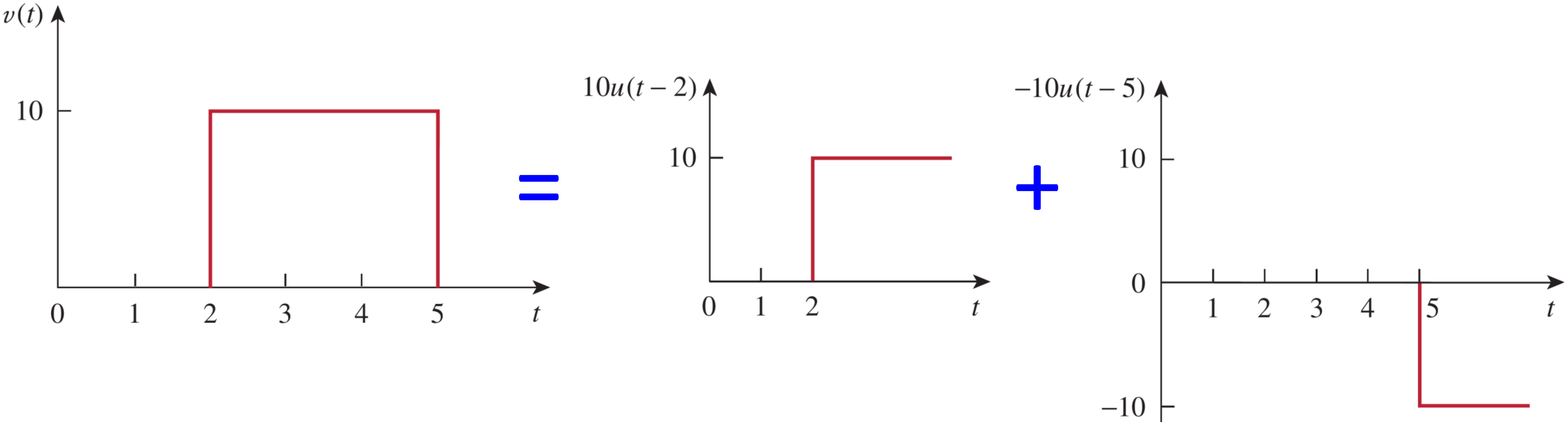
- Unit step function (单位阶跃函数):

a) The unit step function can be used to represent both turning ON and turning OFF.



$$v(t) = 10u(t-2) - 10u(t-5) = 10[u(t-2) - u(t-5)]$$

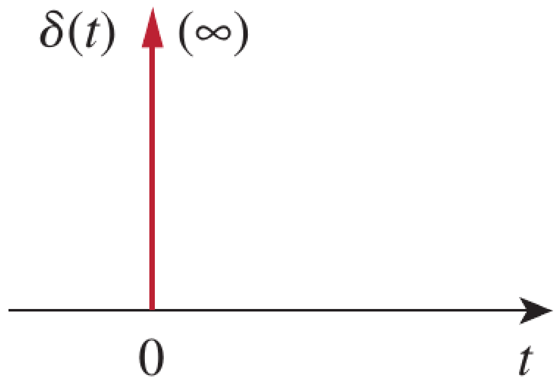
7.6 Singularity Functions (d)

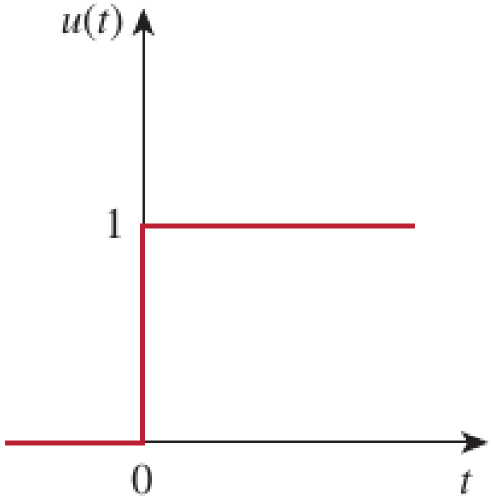
- Unit impulse function (单位冲激函数):

a) The unit impulse function $\delta(t - t_0)$ is zero everywhere except at t_0, where it is undefined.

b) One practical definition of delta function is to define it as the derivative of the unit step function $u(t)$.

c) Impulsive current or voltage is good model of the discharging current when a charged capacitor shorted or the voltage of a current-carrying inductor is opened.





$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$

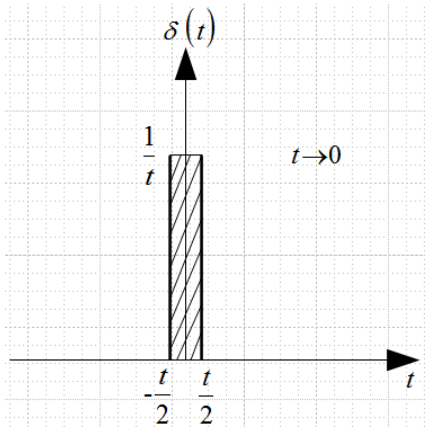
7.6 Singularity Functions (e)

- The strength of the unit impulse function:

a) The unit impulse function $\delta(t - t_0)$ is zero everywhere except at t_0, where it is undefined.

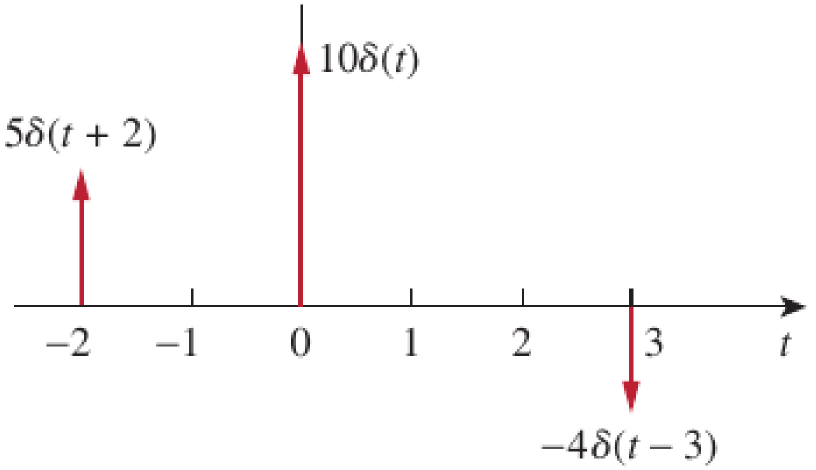
b) The unit area is known as the strength of the impulse function. When an impulse function has a strength other than unity, the area of the impulse is equal to its strength.

c) Impulsive current or voltage is good model of the discharging current when a charged capacitor shorted or the voltage of a current-carrying inductor is opened.



$$\int_{0^-}^{0^+} \delta(t)\, dt = 1$$

$$\int_a^b f(t)\delta(t - t_0)\, dt = f(t_0)$$



7.6 Singularity Functions (f)

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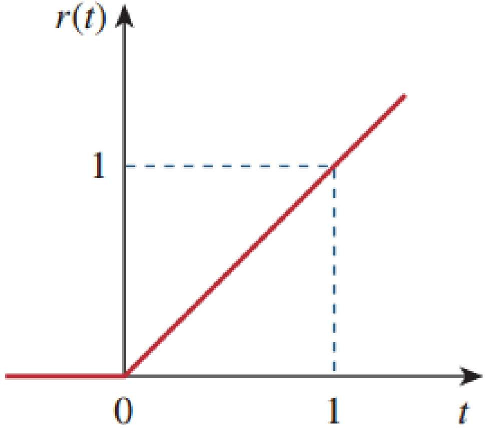
- The unit ramp function:



$$r(t)=\int_{-\infty}^{t}u(\tau)d\tau=tu(t)=\begin{cases}0, & t<0\\ t, & t\geq 0\end{cases}$$

Voltage response of capacitor to constant current input
Current response of inductor to constant voltage input

The relationship between impulse, step, and ramp functions

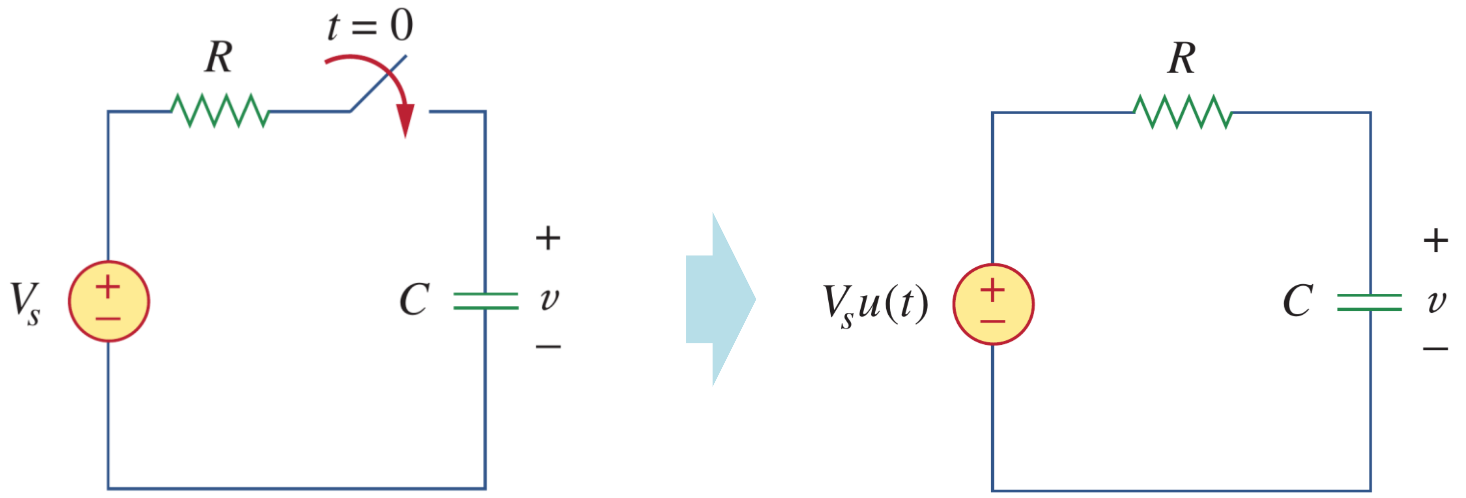
In form of differentiation: $\boxed{\delta(t)=\dfrac{du(t)}{dt},\ u(t)=\dfrac{dr(t)}{dt}}$

In form of integration: $\boxed{u(t)=\int_{-\infty}^{t}\delta(\lambda)d\lambda,\ r(t)=\int_{-\infty}^{t}u(\lambda)d\lambda}$

Lesson 7 First-order circuits

7.7 Step-response of RC circuits (a)

- Step-response of RC circuits :



a) The step response (阶跃响应) of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.

b) In simple words, when RC circuit is connected with a DC source at certain instant $t = t_0$, the applied voltage can be modeled as a step function, and the corresponding response is called step response.

7.7 Step-response of RC circuits (b)

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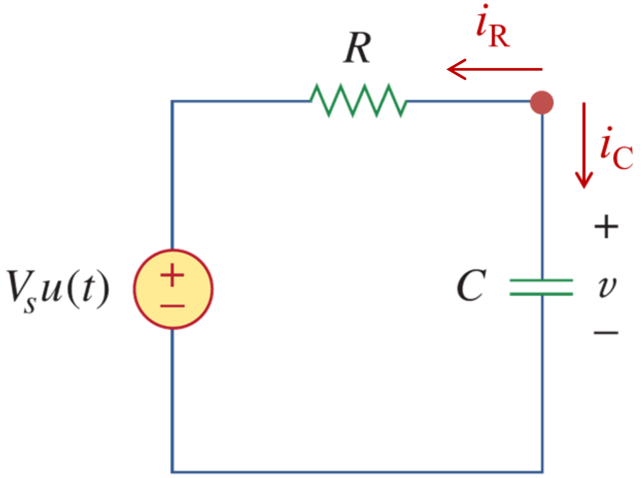
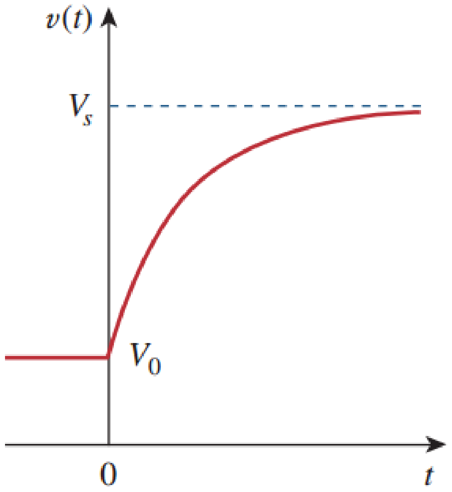
- Step-response of RC circuits :





a) Assume that the initial voltage across capacitor is $v(0) = V_0$. Applying KCL, yielding:

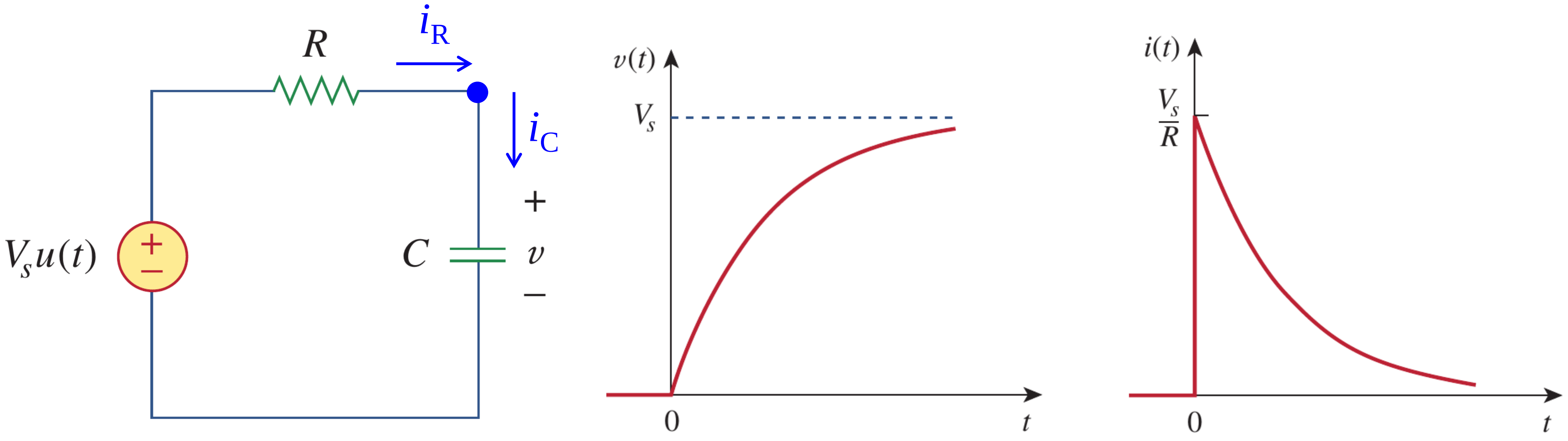
$$C\frac{dv(t)}{dt} + \frac{v(t) - V_s u(t)}{R} = 0$$

b) Being a separable equation (see slide 32), Eq. 1 can be solved as,

$$\boxed{v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}}$$

7.7 Step-response of RC circuits (c)

- Step-response of RC circuits If $V_0 = 0$ V:



If $V_0 = 0$ V,

$$\boxed{v(t) = V_s(1 - e^{-t/\tau})u(t)}$$

$$\boxed{i(t) = C\frac{dv}{dt} = \frac{C}{\tau}V_s e^{-t/\tau} = \frac{V_s}{R}e^{-t/\tau}u(t)}$$

7.7 Step-response of RC circuits (d)

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- Understanding of Step-response of RC circuits:

a) The complete response can be decomposed into two separate responses in two ways: (1) a natural response (自由响应) plus a forced response (受迫响应) or (2) a transient response (暂态响应) plus a steady-state response (稳态响应).

b) Perspective 1:

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau} = V_0 e^{-t/\tau} + V_s(1 - e^{-t/\tau}) = v_n + v_f$$

$$If\ \boldsymbol{v_n = V_0 e^{-t/\tau}}\ and\ \boldsymbol{v_f = V_s(1 - e^{-t/\tau})},$$

$\downarrow$ $\downarrow$

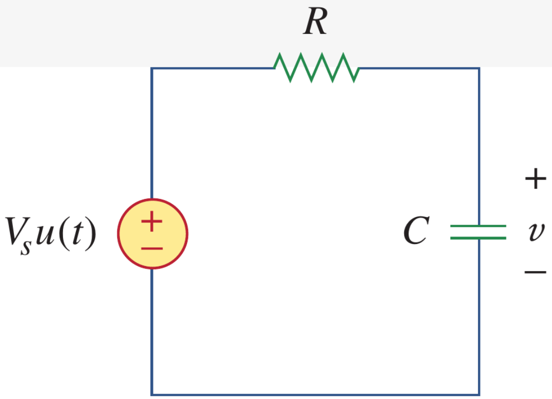
Natural response Stored energy Decaying to zero	+	**Forced response** Independent sources Approaching to V_s	=	**Complete response**

- Understanding of Step-response of RC circuits:

a) The complete response can be decomposed into two separate responses in two ways: (1) a natural response (自由响应) plus a forced response (受迫响应) or (2) a transient response (暂态响应) plus a steady-state response (稳态响应).

b) Perspective 2:

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau} = v_{ss} + v_t$$



$$If\ v_{ss} = V_s\ and\ v_t = (V_0 - V_s)e^{-t/\tau}$$

$\downarrow$ $\downarrow$

Steady-state response Permanent part $v(\infty)$ After all transients are done	+	**Transient response** Temporary part Decaying to zero	=	**Complete response**

7.7 Step-response of RC circuits (f)

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★★★

- General Approach to Solving the RC Circuit:

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a) From these two perspectives, the step response of an RC circuit can be written as (1) $\boldsymbol{v(t) = v_n + v_f = v(0)e^{-t/\tau} + v(\infty)[1 - e^{-t/\tau}]}$

or, (2) $\boldsymbol{v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}}$

where $v(0), v(\infty)$ are the initial and final (or steady-state) voltages.

II

b) Find **- the initial capacitor voltage *v*(0)**
- the final capacitor voltage *v*(∞)
- the time constant τ = *RC*.

III

c) When excitation is applied at $t = t_0$ instead of $t = 0$, the response will be accordingly delayed by t_0

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

where $v(t_0)$ is the initial voltage at $t = t_0^+$.

7.8 Examples (a)

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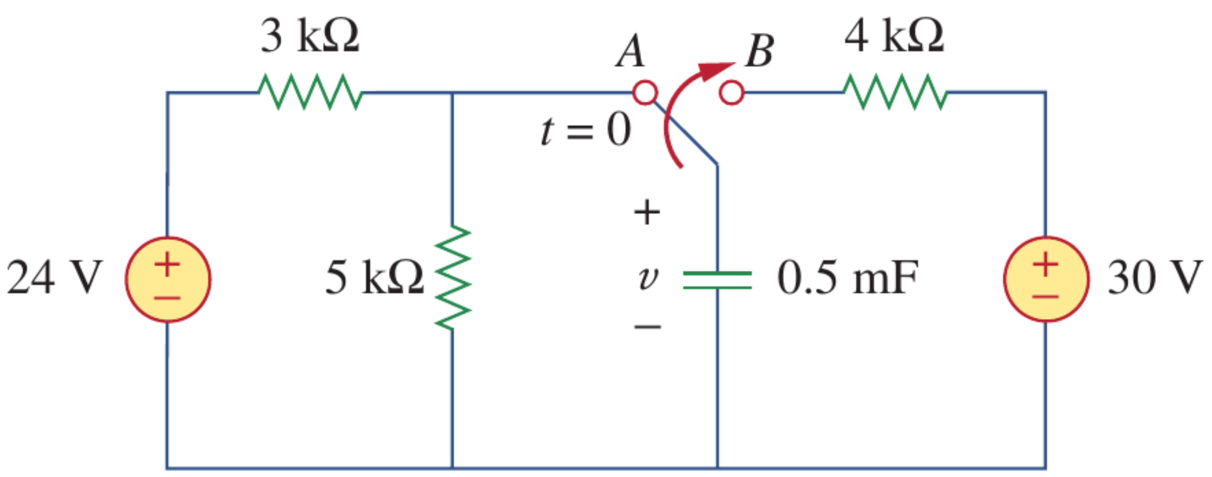
- Example 7.07:

Example 7.07: The switch has been in position A for a long time. At t = 0, the switch moves to B. Determine $v(t)$ for t > 0 and calculate its value at t = 1 s and 4 s.



Solution:

$$v(t) = v_{ss} + v_t = v(\infty) + \left[v(0) - v(\infty)\right]e^{-t/\tau}$$

→ The time constant → The equivalent resistance

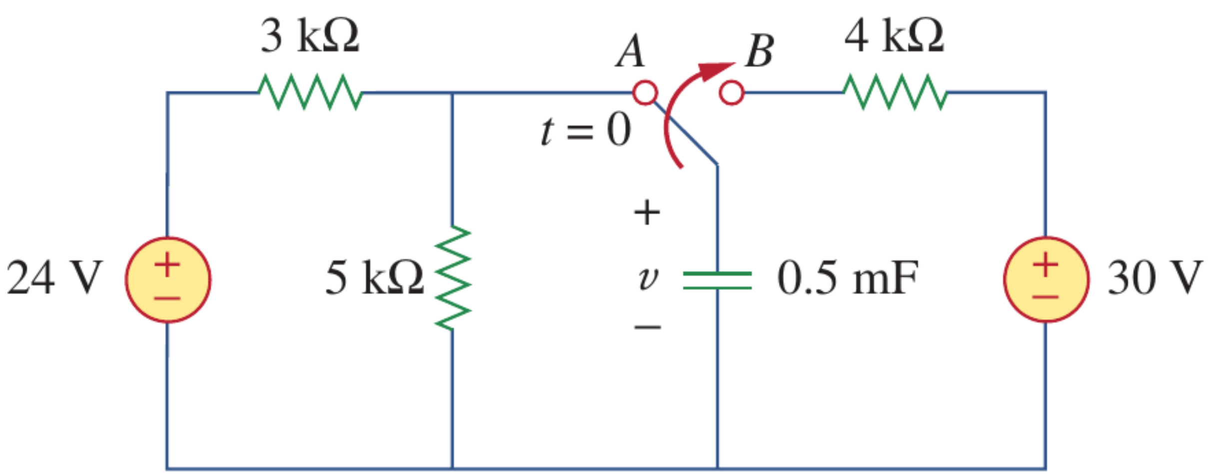
The final steady state voltage

The initial voltage

7.8 Examples (b)

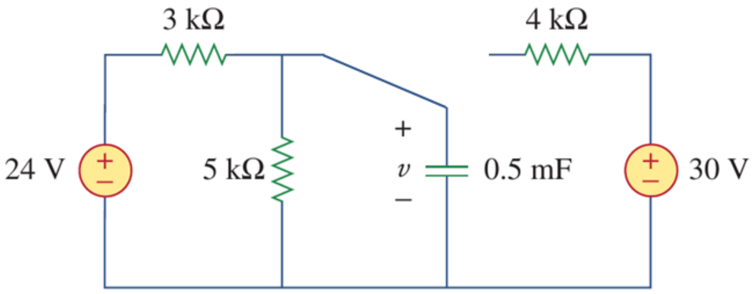
- Example 7.07:

Example 7.07: The switch has been in position A for a long time. At t = 0, the switch moves to B. Determine $v(t)$ for t > 0 and calculate its value at t = 1 s and 4 s.



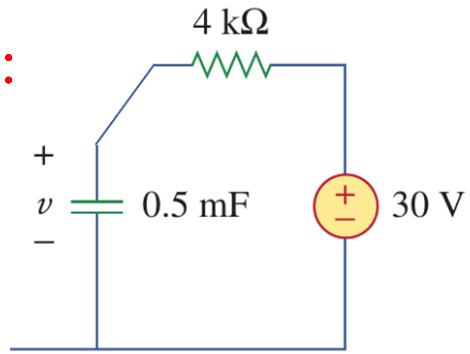
Solution:

(1) $t < 0$ for $v(0)$:



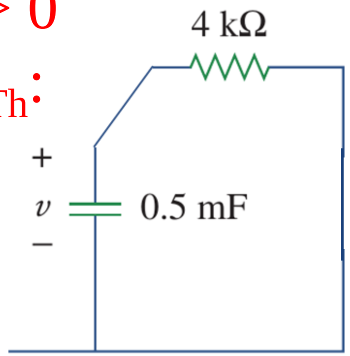
$$v(0) = \frac{5}{3+5} \times 24 = 15\ \mathrm{V}$$

(2) $t > 0$ for $v(\infty)$:



$$v(\infty) = 30\ \mathrm{V}$$

(3) $t > 0$ for R_{Th}:

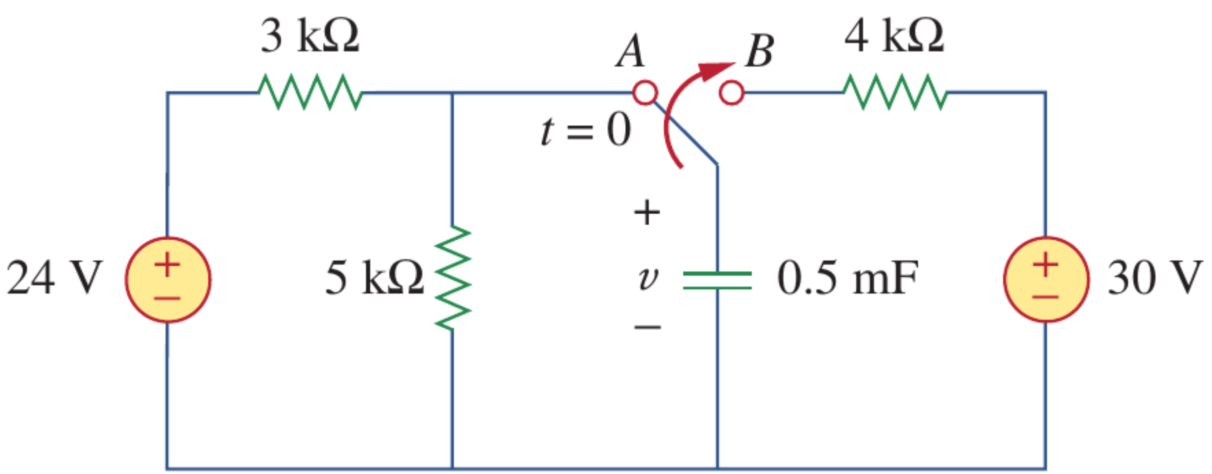


$$R_{eq} = 4\ \mathrm{k\Omega}$$

7.8 Examples (c)

- Example 7.07:

Example 7.07: The switch has been in position A for a long time. At t = 0, the switch moves to B. Determine $v(t)$ for t > 0 and calculate its value at t = 1 s and 4 s.



Solution: $\tau = R_{Th}C = 0.5\,m \times 4\,k = 2\ \mathrm{S}$

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

$$= 30 + [15 - 30]e^{-t/(0.5\mathrm{m} \times 4\mathrm{k})}$$

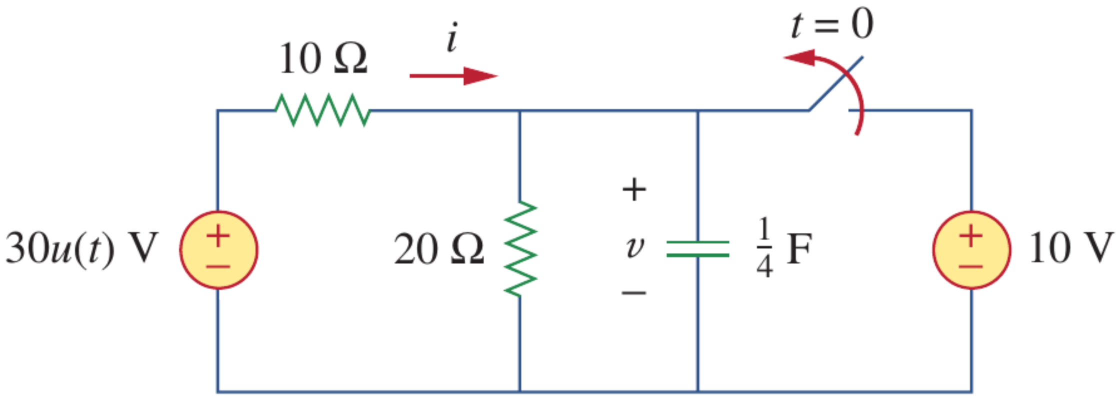
$$= 30 - 15e^{-t/2}$$

$$v_1 = 30 - 15e^{-0.5} = 20.9\ \mathrm{V} \quad v_2 = 30 - 15e^{-2} = 27.97\ \mathrm{V}$$

7.8 Examples (d)

- Example 7.08:

Example 7.08: In the following circuit, the switch has been closed for a long time and is opened at t = 0. Find $i(t)$ and $v(t)$ for $t > 0$.



Solution:

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

The final steady state voltage

The initial voltage

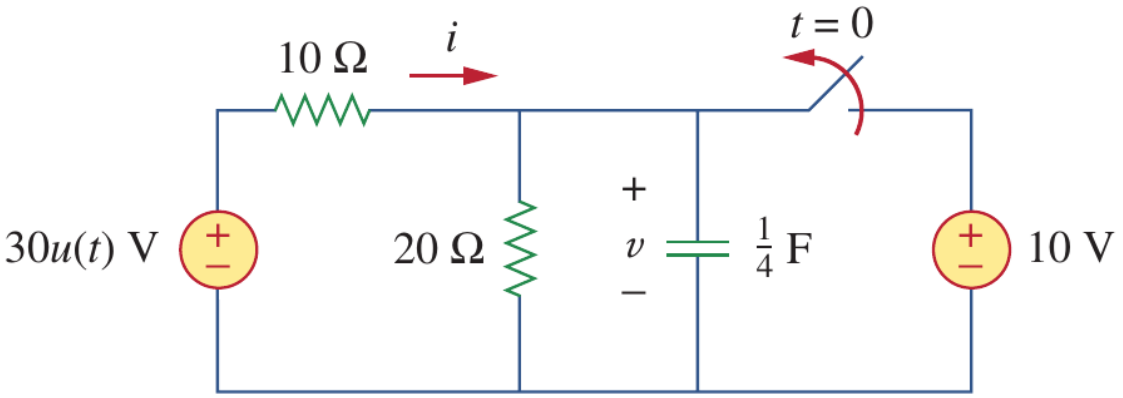
The time constant

The equivalent resistance

7.8 Examples (d)

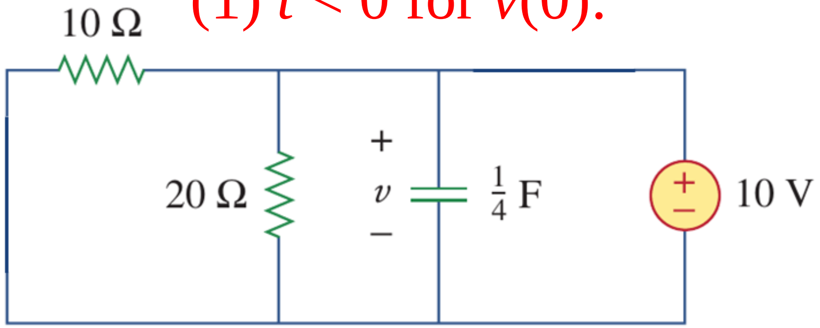
- Example 7.08:

Example 7.08: In the following circuit, the switch has been closed for a long time and is opened at $t = 0$. Find $i(t)$ and $v(t)$ for $t > 0$.



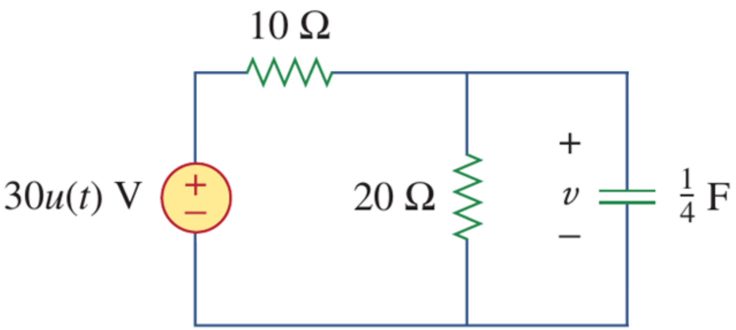
Solution:

(1) $t < 0$ for $v(0)$:



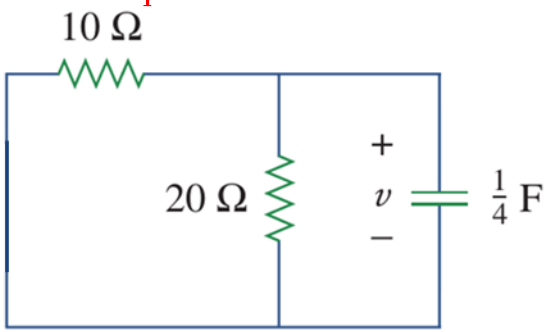
$$v(0) = 10\text{ V}$$

(2) $t > 0$ for $v(\infty)$:



$$v(\infty) = \frac{20}{10+20} \times 30 = 20\text{ V}$$

(3) $t > 0$ for R_{eq}:



$$R_{eq} = \frac{20 \times 10}{10+20} = \frac{20}{3}\ \Omega$$

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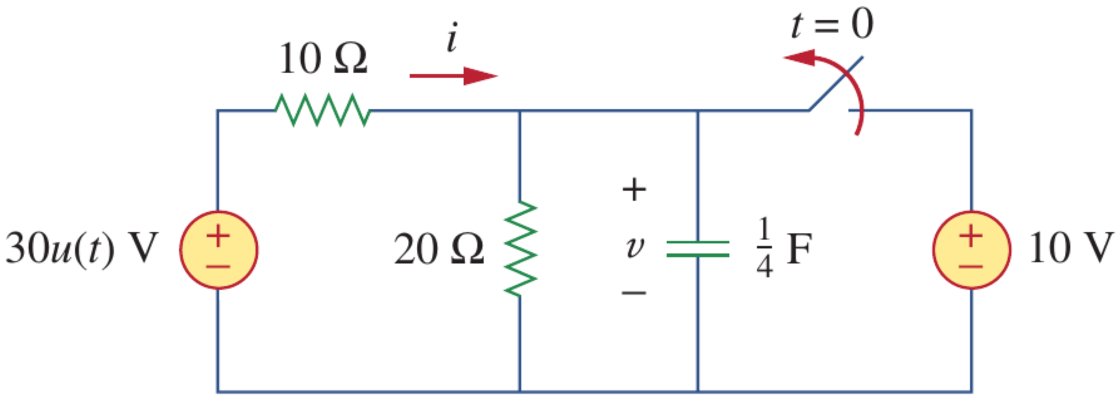
- Example 7.08:

Example 7.08: In the following circuit, the switch has been closed for a long time and is opened at $t = 0$. Find $i(t)$ and $v(t)$ for $t > 0$.



Solution:

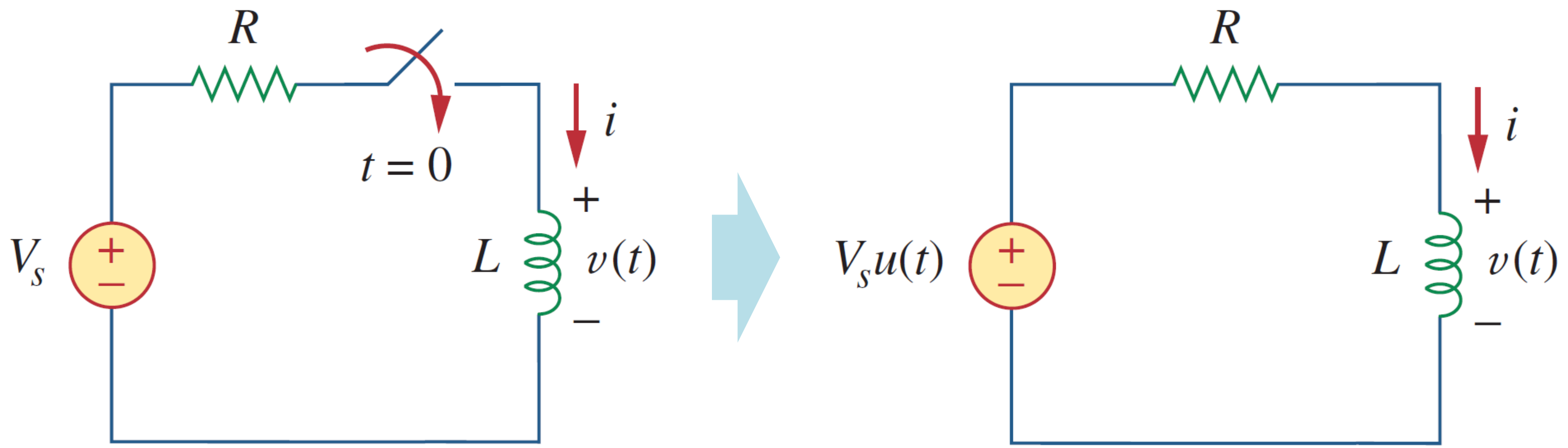
$$\tau = R_{Th}C = \frac{20}{3} \times \frac{1}{4} = \frac{5}{3}\ \mathrm{S}$$

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$
$$= 20 + [10 - 20]e^{-3t/5}$$
$$= 20 - 10e^{-0.6t}$$

$$i(t) = \frac{v}{20} + C\frac{dv}{dt} = 1 + e^{-0.6t}$$

7.9 Step-response of RL circuits (a)

- Step-response of RL circuits :



a) The step response (阶跃响应) of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.

b) In simple words, when an RL circuit is connected with a DC source at certain instant $t = t_0$, the applied voltage can be modeled as a step function, and the corresponding response is called step response.

7.9 Step-response of RL circuits (b)

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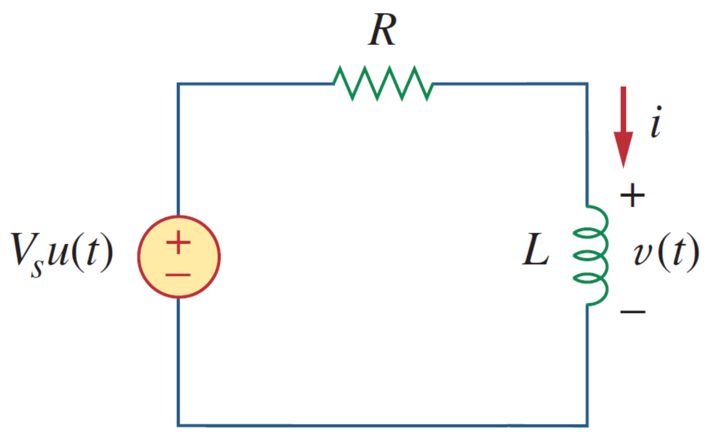
- Step-response of RL circuits :



a) Assume that the initial voltage across capacitor is $v(0) = V_0$. Applying KVL, yielding:

$$v_s = Ri(t) + L\frac{di(t)}{dt}$$

b) Being a separable equation (see slide 32), Eq. 1 can be solved as,

$$\boxed{i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-t/\tau}}$$

7.9 Step-response of RL circuits (c)

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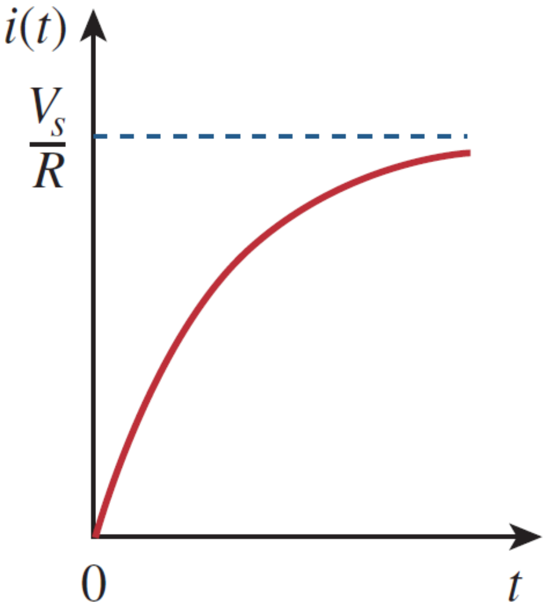
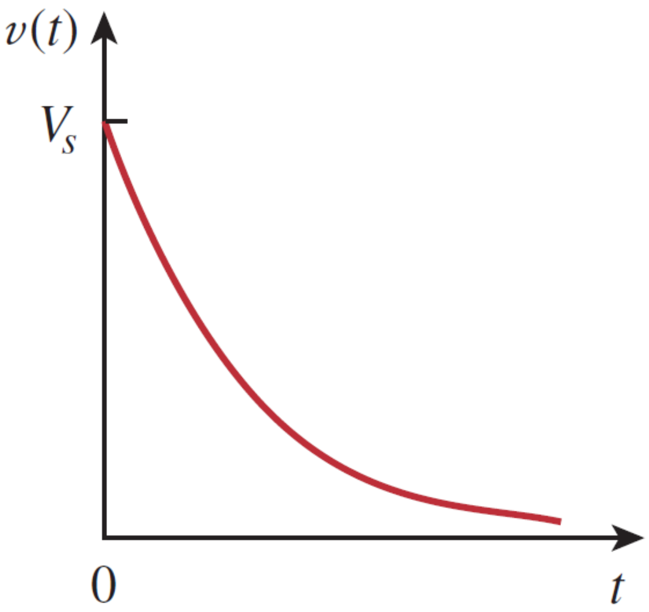
- Step-response of RL circuits If $I_0 = 0$ A:





If $I_0 = 0$ V,

$$\boxed{i(t) = \frac{V_s}{R}(1 - e^{-t/\tau})u(t)}$$

$$\boxed{v(t) = V_s e^{-t/\tau}u(t), \qquad \tau = \frac{L}{R}}$$

7.9 Step-response of RL circuits (d)

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- Understanding of Step-response of RL circuits:

a) The complete response can be decomposed into two separate responses in two ways: (1) a natural response (自由响应) plus a forced response (受迫响应) or (2) a transient response (暂态响应) plus a steady-state response (稳态响应).

b) Perspective 1:

$$i(t) = \frac{v_s}{R} + \left(I_0 - \frac{v_s}{R}\right)e^{-t/\tau} = I_0 e^{-t/\tau} + \frac{v_s}{R}\left(1 - e^{-t/\tau}\right) = i_n + i_f$$

$$If\ i_n = I_0 e^{-t/\tau}\ and\ i_f = \frac{v_s}{R}\left(1 - e^{-t/\tau}\right),$$

$\downarrow$ $\downarrow$

Natural response Stored energy Decaying to zero	+	**Forced response** Independent sources Approaching to V_s/R	=	**Complete response**

7.9 Step-response of RL circuits (e)

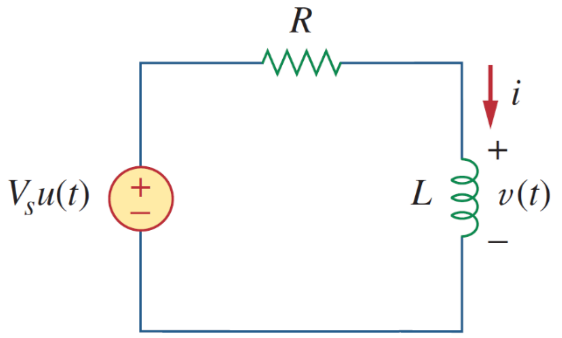
- Understanding of Step-response of RL circuits:

a) The complete response can be decomposed into two separate responses in two ways: (1) a natural response (自由响应) plus a forced response (受迫响应) or (2) a transient response (暂态响应) plus a steady-state response (稳态响应).

b) Perspective 2:

$$i(t) = \frac{v_s}{R} + (I_0 - \frac{v_s}{R})e^{-t/\tau} = i_{ss} + i_t$$

$$If\ i_{ss} = \frac{v_s}{R}\ and\ i_t = (I_0 - \frac{v_s}{R})e^{-t/\tau}$$



Steady-state response
Permanent part ***i(∞)***
After all transients are done

+

Transient response
Temporary part
Decaying to zero

=

Complete response

7.9 Step-response of RL circuits (f)

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- **General Approach to Solving the RL Circuit:** ★★★

I

a) From these two perspectives, the step response of an RL circuit can be written as (1) $\boldsymbol{i(t) = i_n + i_f = i(0)e^{-t/\tau} + i(\infty)[1 - e^{-t/\tau}]}$

or, (2) $\boldsymbol{i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}}$

where $i(0), i(\infty)$ are the initial and final (or steady-state) currents.

II

b) Find **- the initial inductor current *i*(0)**
- the final inductor voltage *i*(∞)
- the time constant τ = *L*/*R*.

III

c) When excitation is applied at $t = t_0$ instead of $t = 0$, the response will be accordingly delayed by t_0

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

where $i(t_0)$ is the initial voltage at $t = t_0^+$.

7.10 Examples (a)

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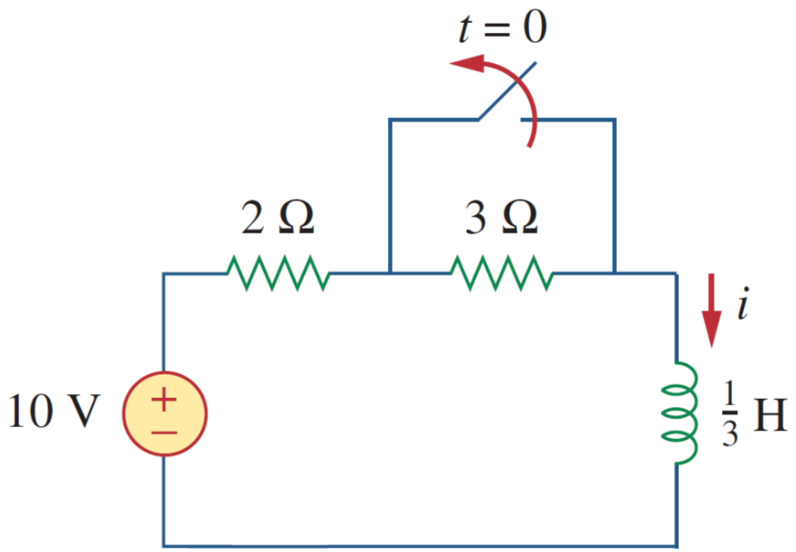
- Example 7.09:

Example 7.09: Find *i(t)* in the following circuit. Assume that the switch has been closed for a long time.



$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

The final steady state current

The initial current

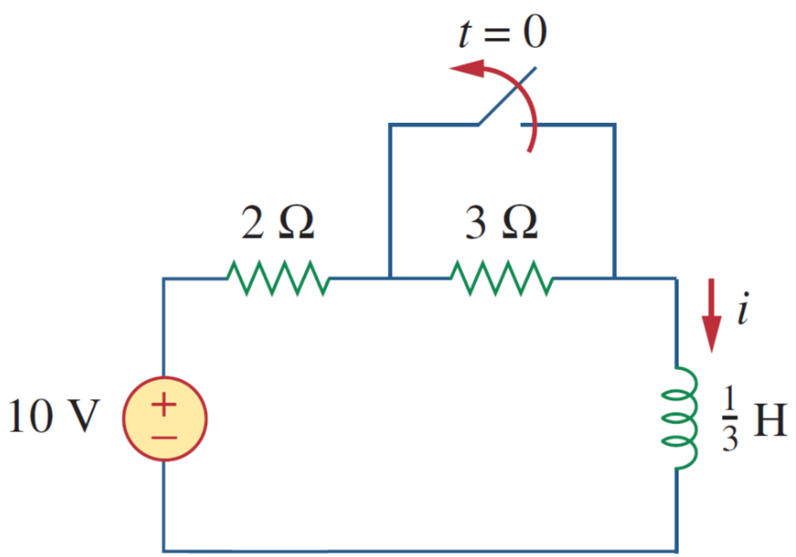
The time constant

The equivalent resistance

7.10 Examples (b)

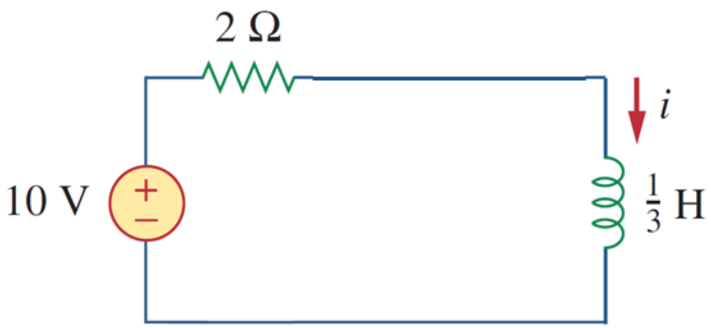
- Example 7.09:

Example 7.09: Find $i(t)$ in the following circuit. Assume that the switch has been closed for a long time.



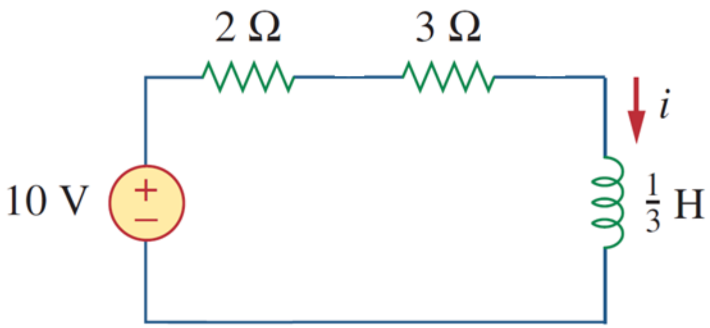
Solution:

(1) $t < 0$ for $i(0)$:



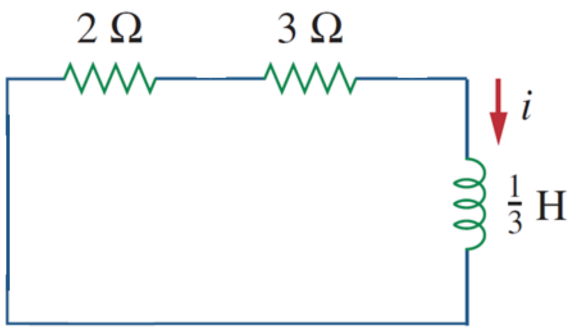
$$i(0) = \frac{10}{2} = 5 \text{ A}$$

(2) $t > 0$ for $i(\infty)$:



$$i(\infty) = \frac{10}{2+3} = 2 \text{ A}$$

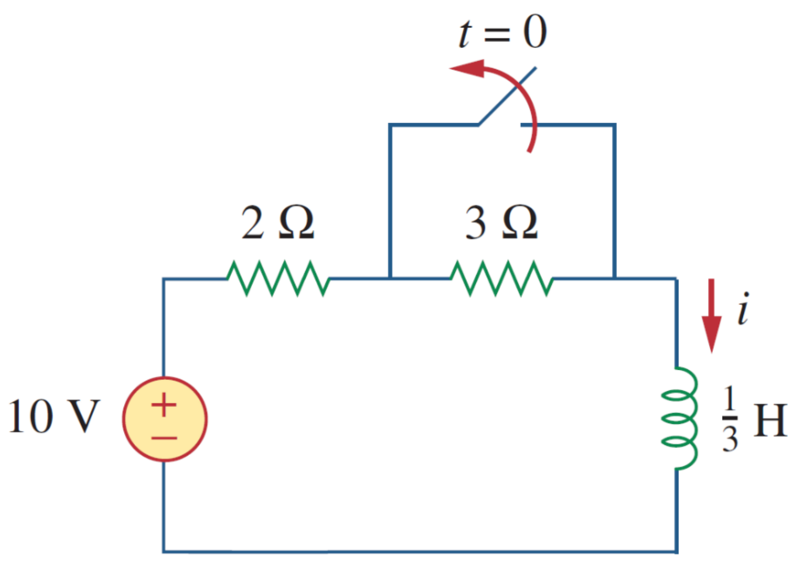
(3) $t > 0$ for R_{eq}):



$$R_{eq} = 2 + 3 = 5\ \Omega$$

7.10 Examples (c)

- Example 7.09:



Example 7.09: Find *i(t)* in the following circuit. Assume that the switch has been closed for a long time.

Solution:

(4) The time constant: $\tau = \frac{L}{R_{\mathrm{Th}}} = \frac{\frac{1}{3}}{5} = \frac{1}{15}\ \mathrm{s}$

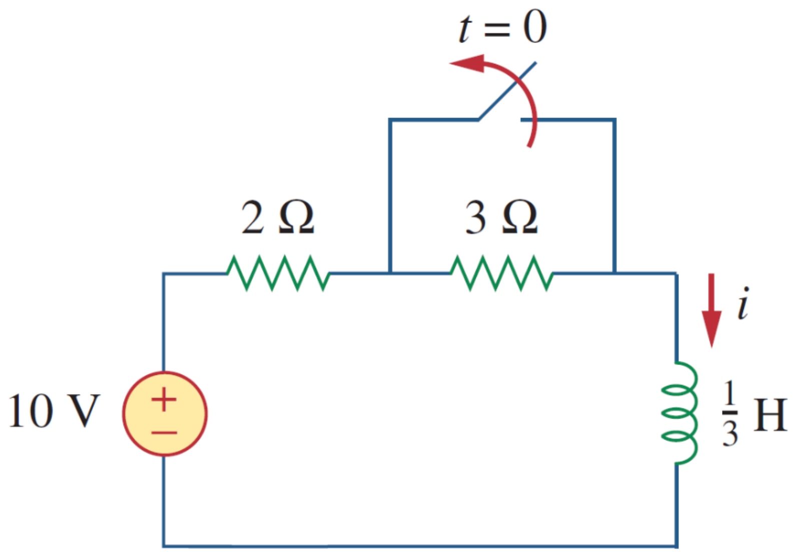
(5) The current:

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$
$$= 2 + (5-2)e^{-15t} = 2 + 3e^{-15t}\ \mathrm{A}, \qquad t > 0$$

7.10 Examples (d)

- Example 7.09:

Example 7.09: Find *i(t)* in the following circuit. Assume that the switch has been closed for a long time.



Solution:

(6) Check your results using KVL for the loop:

$$10 = 5i + L\frac{di}{dt}$$

$$5i + L\frac{di}{dt} = [10 + 15e^{-15t}] + \left[\frac{1}{3}(3)(-15)e^{-15t}\right] = 10$$

Therefore the results are confirmed.

7.10 Examples (e)

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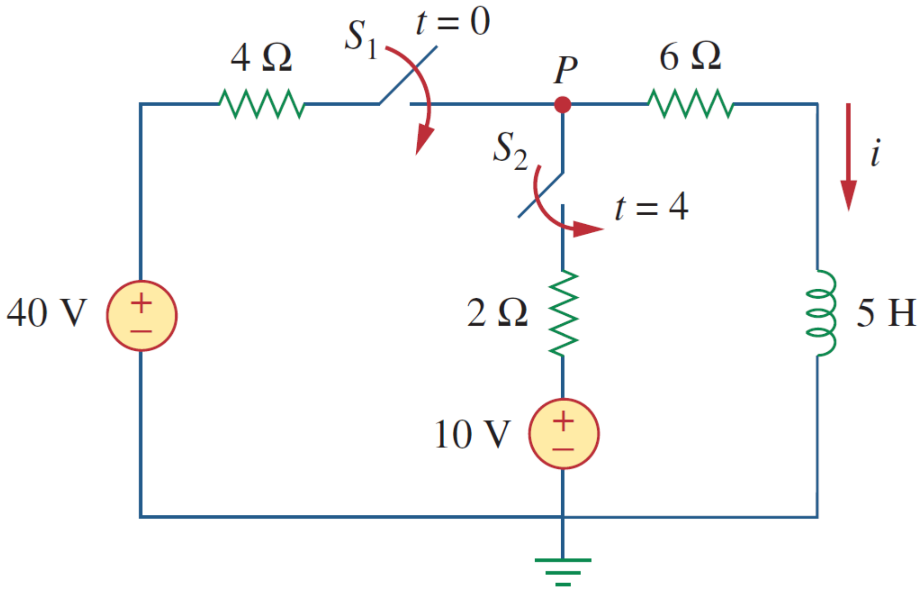
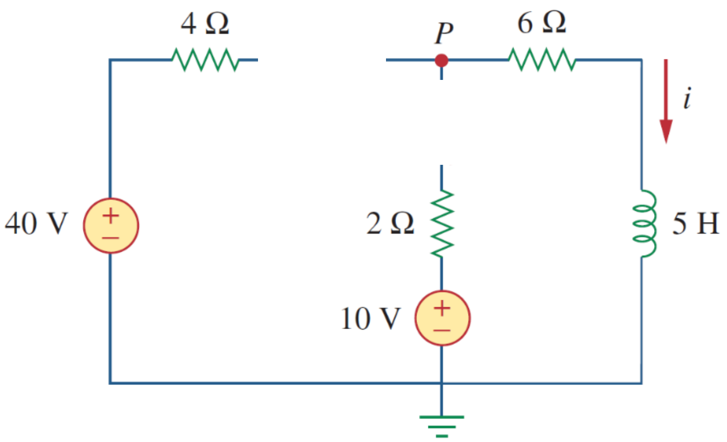
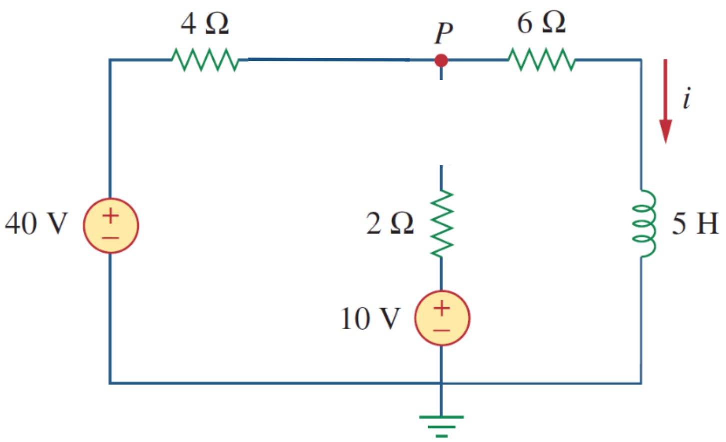
- Example 7.10:



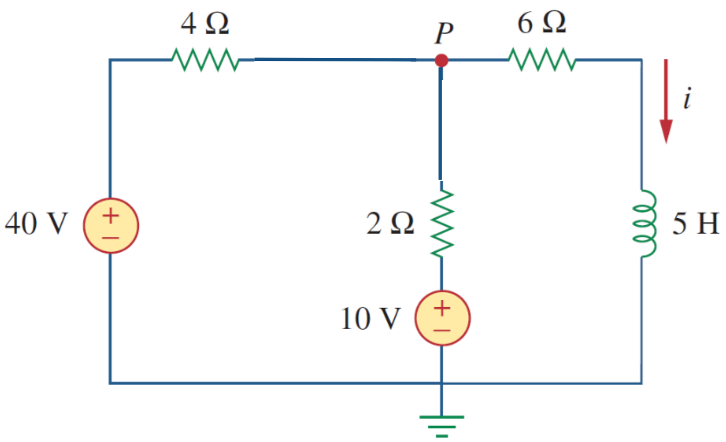
Example 7.10: Switches 1 and 2 have been open for a long time. At $t = 0$, switch 1 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for t > 0. Calculate i for t = 2 s and t = 5 s.



$t < 0$



$0 < t \leq 4$

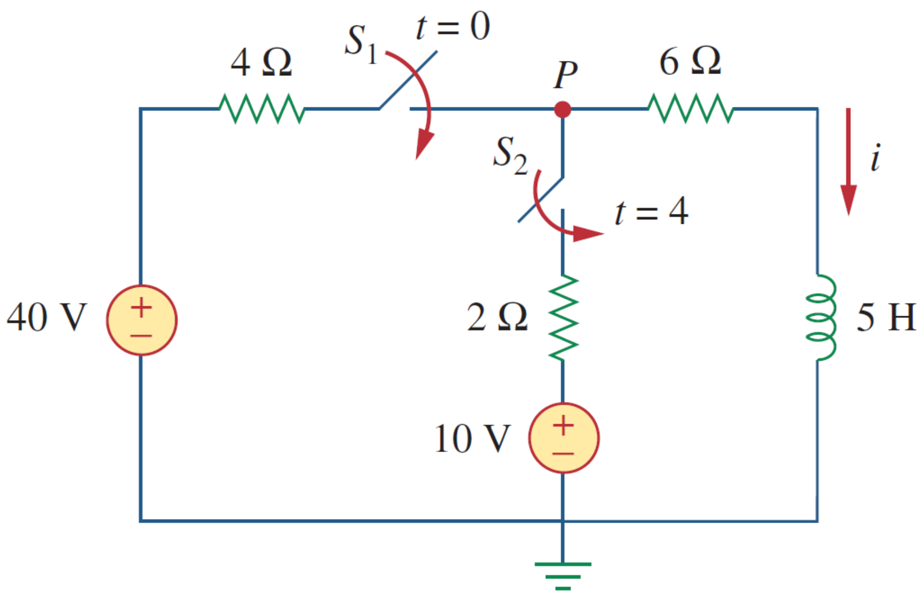


$t > 4$

7.10 Examples (f)

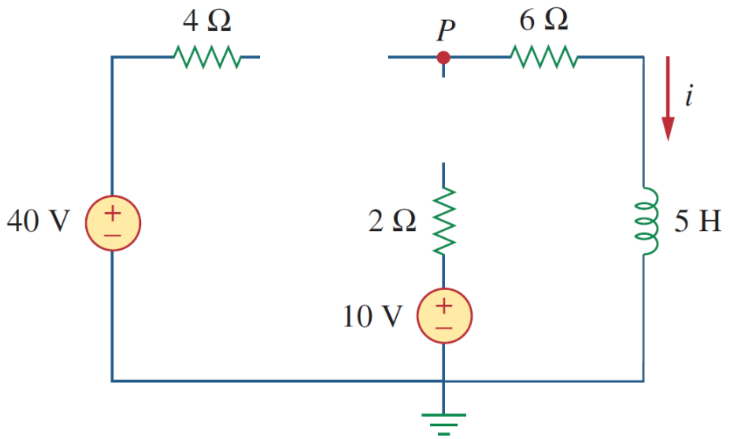
- Example 7.10:

Example 7.10: Switches 1 and 2 have been open for a long time. At $t = 0$, switch 1 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for t > 0. Calculate i for t = 2 s and t = 5 s.



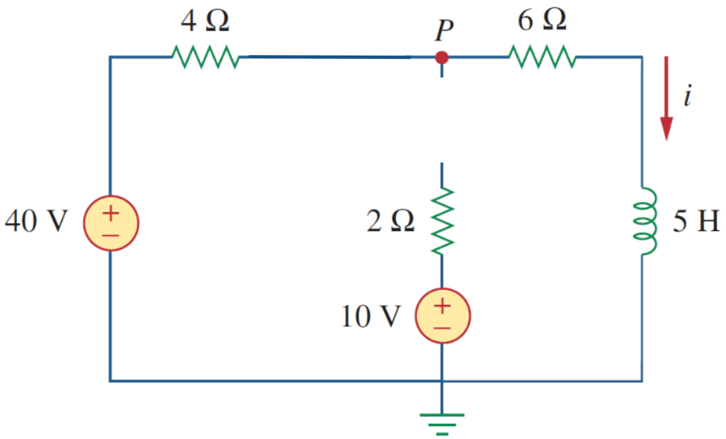
For "S1 turning ON"

(1) $t < 0$: $\boldsymbol{i(0) = 0}$ **A**



(2) $0 < t \le 4$:

$\boldsymbol{i(\infty)}$ **= 4 A and** $\boldsymbol{R_{Th}}$ **= 10 Ω**



(3) $\tau = L/R_{Th} = 0.5$ s

$$i = \begin{cases} 0\ \mathrm{A}, t < 0 \\ 4(1 - e^{-2t})\ \mathrm{A}, 0 < \mathrm{t} \le 4 \end{cases}$$

$i \approx 3.93$ A at $t = 2$ s

$i \approx 4$ A at $t = 4$ s

7.10 Examples (g)

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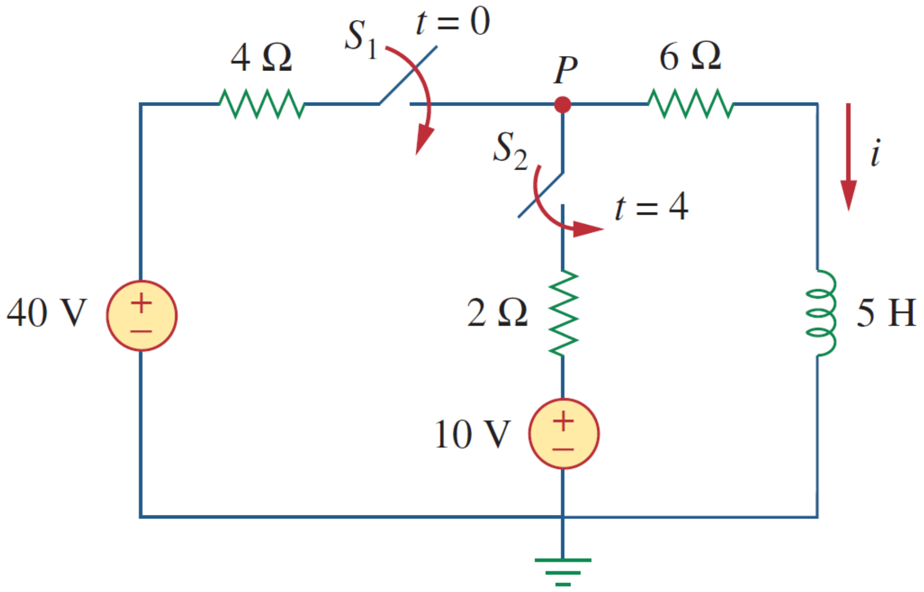
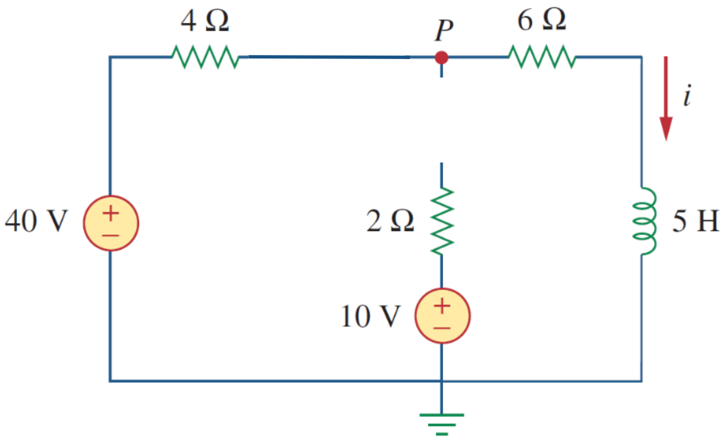
- Example 7.10:

Example 7.10: Switches 1 and 2 have been open for a long time. At t = 0, switch 1 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for t > 0. Calculate i for t = 2 s and t = 5 s.



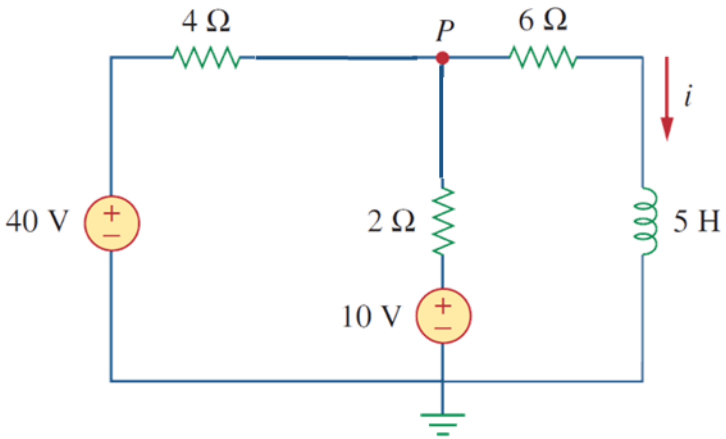
For "S2 turning ON"

(5) $t = 4$: $\mathbf{i'(0) = i(4) = 4\ A}$



(6) $t > 4$:

$\mathbf{i'(\infty) = ??\ A}$ **and** $\mathbf{R'_{Th} = 22/3\ \Omega}$



Using nodal analysis:

$$\frac{40 - v}{4} + \frac{10 - v}{2} = \frac{v}{6}$$

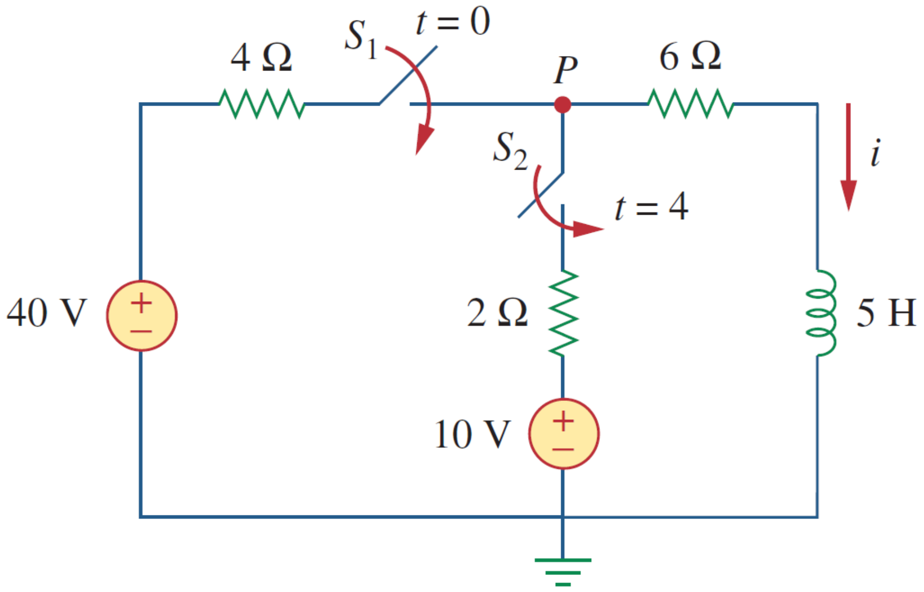
v = 180/11 V

$i'(\infty) = v/6 = 30/11 = 2.727$ A

7.10 Examples (h)

- Example 7.10:

Example 7.10: Switches 1 and 2 have been open for a long time. At t = 0, switch 1 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for t > 0. Calculate i for t = 2 s and t = 5 s.



For "S2 turning ON" $t > 4$

$$\tau = \frac{L}{R_{Th}} = \frac{5}{\frac{22}{3}} = \frac{15}{22} \text{ s}$$

$$i(t) = i(\infty) + [i(4) - i(\infty)]e^{-(t-4)/\tau}, \qquad t \geq 4$$

We need $(t - 4)$ in the exponential because of the time delay. Thus,

$$i(t) = 2.727 + (4 - 2.727)e^{-(t-4)/\tau}, \qquad \tau = \frac{15}{22}$$

$$= 2.727 + 1.273e^{-1.4667(t-4)}, \qquad t \geq 4$$

$$i(5) = 2.727 + 1.273e^{-1.4667}$$

$$= 3.02 \text{ A}$$

7.10 Examples (i)

- Responses of RC and RL circuits:

RC circuits:

$$v(t) = v_n + v_f = v(0)e^{-t/\tau} + v(\infty)[1 - e^{-t/\tau}]$$

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

RL circuits:

$$i(t) = i_n + i_f = i(0)e^{-t/\tau} + i(\infty)[1 - e^{-t/\tau}]$$

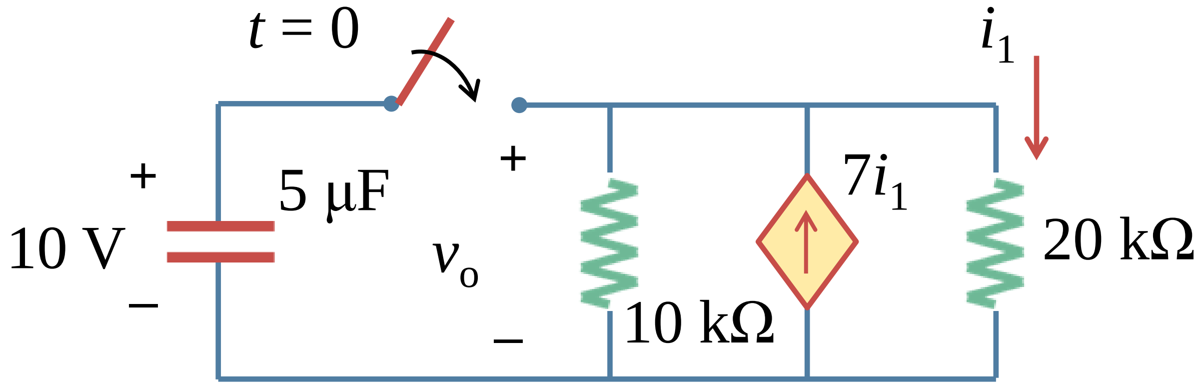
$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

Lesson 7 First-order circuits

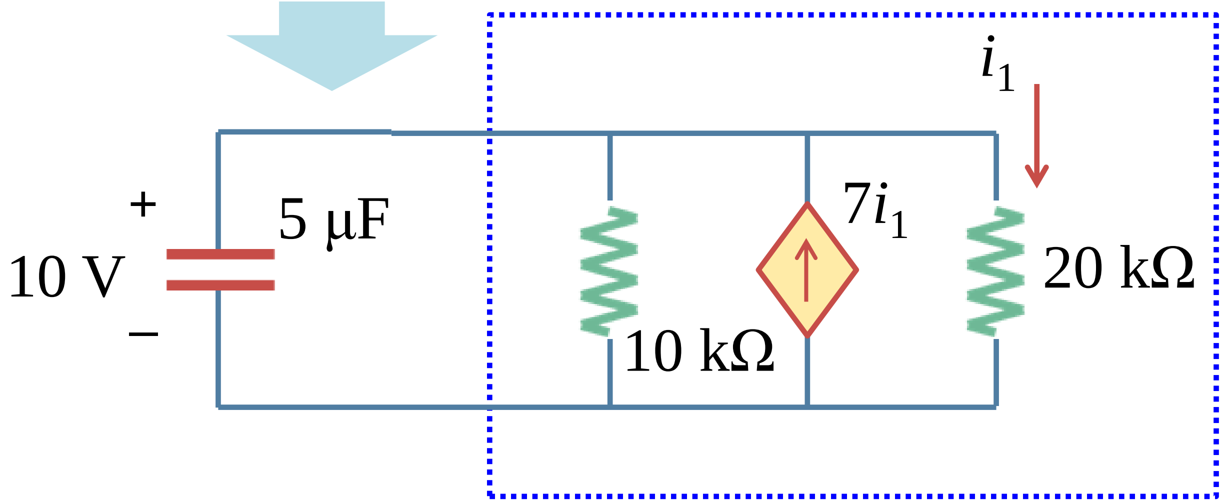
7.11 Unbounded response (a)

- Example 7.11:

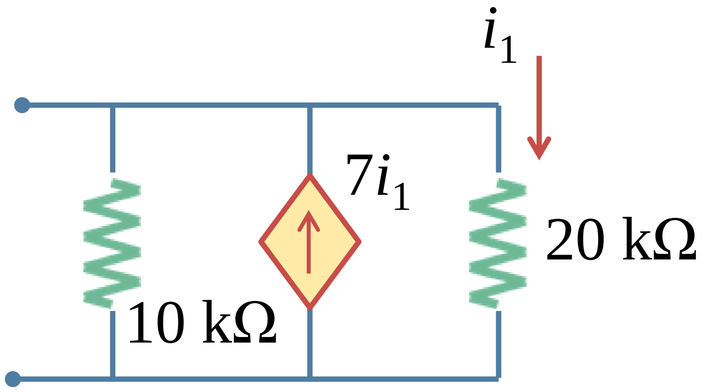
Example 7.11: At t = 0 the switch is closed, and the voltage across the capacitor is 10 V at this moment. Find $v_o(t)$ when $t > 0$.



Solution:



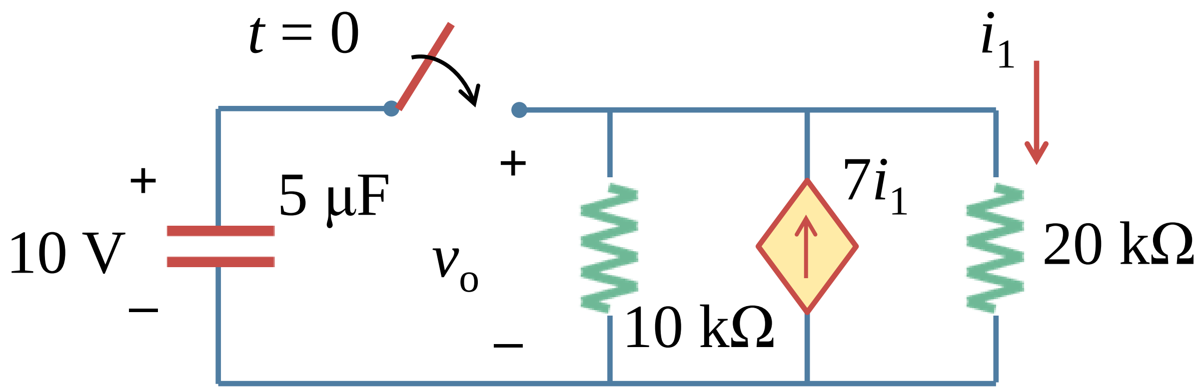
? R_{Th}



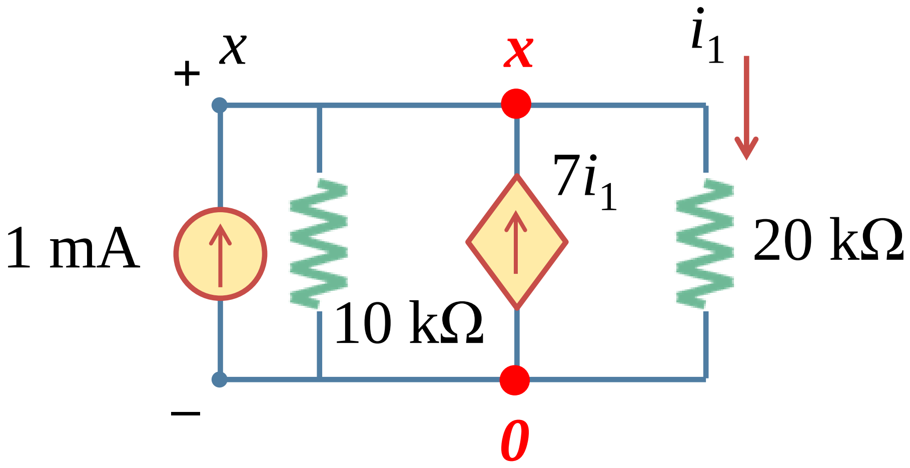
7.11 Unbounded response (b)

- Example 7.11:

Example 7.11: At t = 0 the switch is closed, and the voltage across the capacitor is 10 V at this moment. Find $v_o(t)$ when $t > 0$.



Solution:



Using nodal analysis:

$$0 = 1 - \frac{x}{10} + \frac{7x}{20} - \frac{x}{20} \Rightarrow x = -5 \text{ V}$$

$$R_{Th} = \frac{x}{1\,m} = -5 \text{ k}\Omega$$

7.11 Unbounded response (c)

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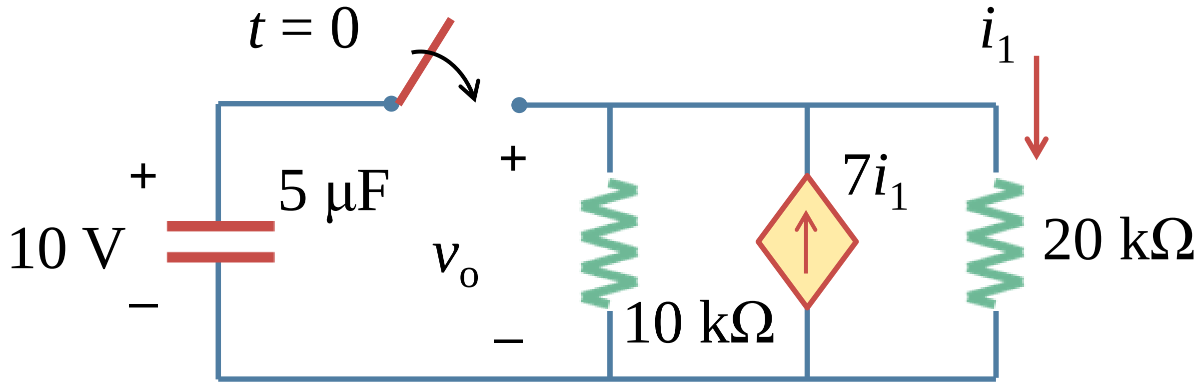
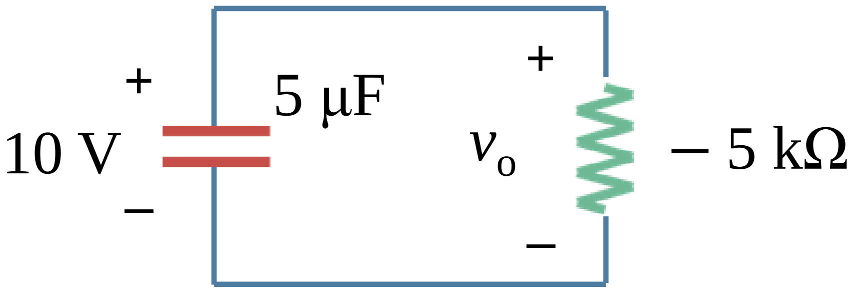
- Example 7.11:

Example 7.11: At t = 0 the switch is closed, and the voltage across the capacitor is 10 V at this moment. Find $v_o(t)$ when t > 0.



Solution:



$$\tau = R_{Th}C = -5\text{ k} \times 5\ \mu = -25\text{ mS}$$

$$v_o(t) = 10 \times e^{-t/\tau} = 10e^{40t}$$

7.11 Unbounded response (d)

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- Unbounded response:

a) A circuit response may grow, rather than decay, exponentially with time. This type of response, called an unbounded response, is possible if the circuit contains dependent sources.

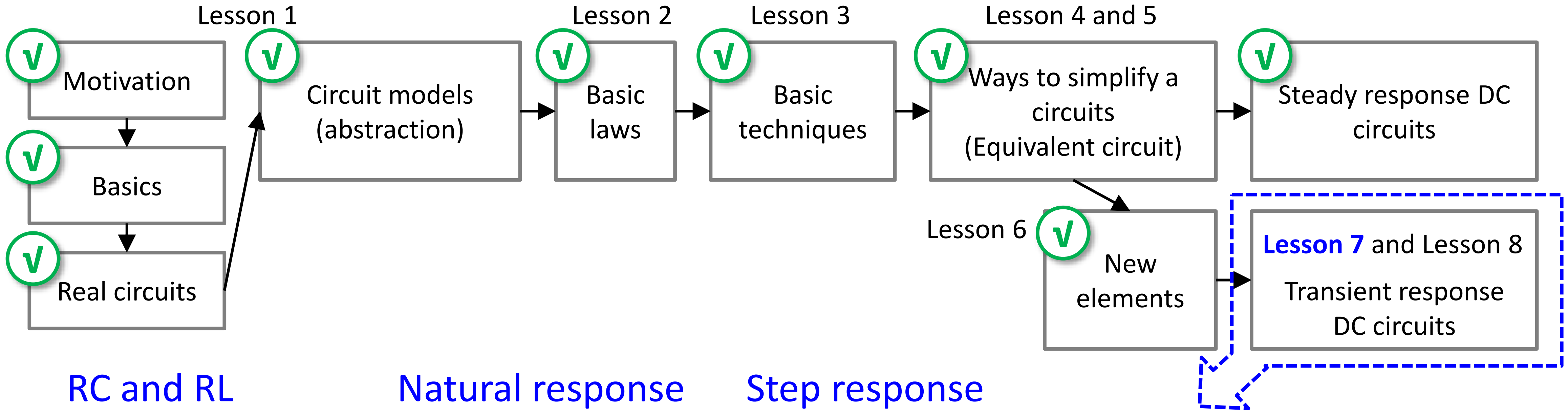
b) In that case, the Thevenin equivalent resistance with respect to the terminals of either an inductor or a capacitor may be negative. This negative resistance generates a negative time constant, and the resulting currents and voltages increase without limit.

c) In an actual circuit, the response eventually reaches a limiting value when a component breaks down or goes into a saturation state, prohibiting further increases in voltage or current.

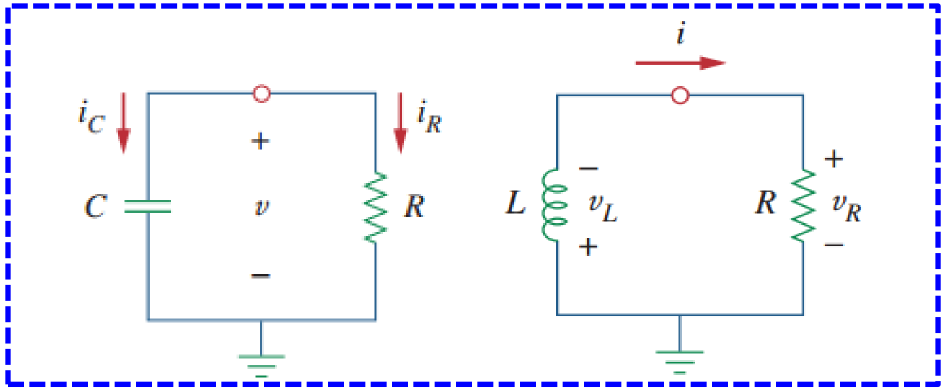
Lesson 7 First-order circuits

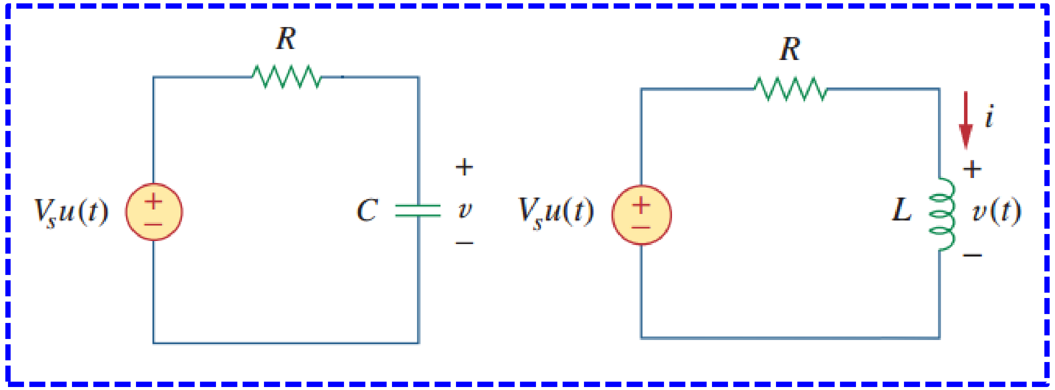
7.13 Summary and assignments (a)

- EE104 road map



RC and RL　　Natural response　　Step response





7.13 Summary and assignments (b)

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- Initial values, final values, time constants !

- The transient response is the component of the complete response that dies out with time.
- The step response is the response of the circuit to a sudden application of a dc current or voltage. Finding the step response of a first-order circuit requires (1) the initial value $x(0^+)$, (2) the final value $x(\infty)$, and (3) the time constant τ. With these three items, we obtain the step response as

$$\boxed{x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}}$$

When the switching occurs at t_0, a more general form of the above equation is

$$\boxed{x(t) = x(\infty) + [x(t_0^+) - x(\infty)]e^{-(t-t_0)/\tau}}$$

Or we may write it as

$$\boxed{\text{Instantaneous value} = \text{Final} + [\text{Initial} - \text{Final}]e^{-(t-t_0)/\tau}}$$

- *You are encouraged to work on the Review Questions to get better understandings.*

7.13 Summary and assignments (c)

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- Important concepts

- The analysis in this chapter is applicable to any circuit that can be reduced to an equivalent circuit comprising a resistor and a single energy-storage element (inductor or capacitor).
  - Such a circuit is of first-order because its behavior is described by a first-order differential equation.
  - In the steady-state dc conditions, the capacitor is an open circuit while the inductor is a shorted circuit.
- The natural response occurs when no independent source is present. It has the general form where x represents current through or voltage across a resistor, a capacitor, or an inductor; x(0) is the initial value of x at t=0.
  - Because most practical resistors, capacitors, and inductors always have losses, the natural response is a transient response, i.e. it dies out with time.
  - The time constant τ is the time required for a response to decay to 1/e of its initial value. For RC circuits, $\tau=RC$ and for RL circuits, $\tau=L/R$.
- The total or complete response consists of the steady-state response and the transient response.
- The steady-state response is the behavior of the circuit after an independent source has been applied for a long time.

7.13 Summary and assignments (d)

- Important concepts

Chapter 7:

Problems 7.7, 7.16, 7.19, 7.29, 7.43, 7.57

(a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.

(b) There are plenty of problems in the textbook, try to complete as many as you can.

Cirrus clouds